

Bioinformatics (COMP0082): Experimental Techniques in -omics and systems biology

Dr D Buchan

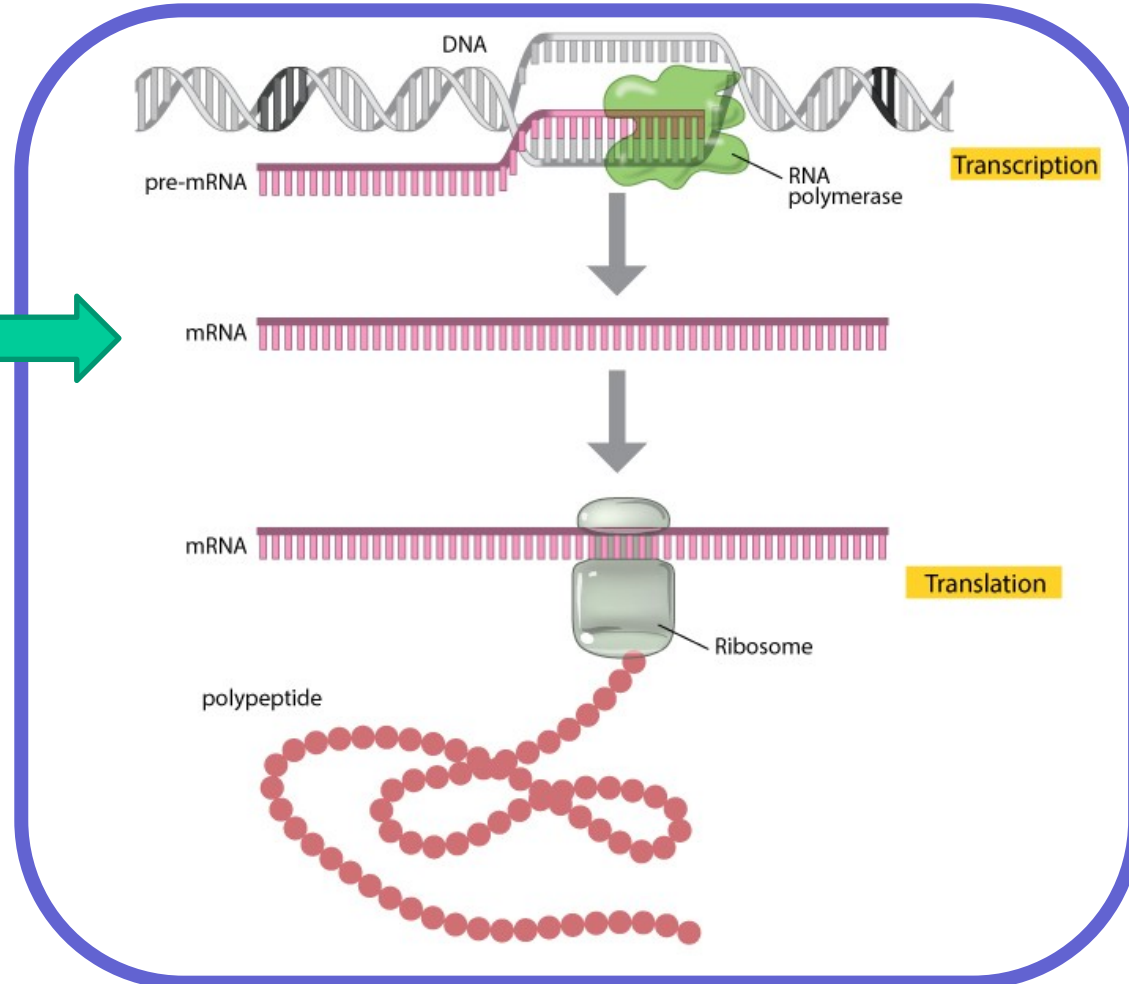
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Topics

1. Transcriptomics Experiments
2. Proteomics Experiments
3. Metabolomics Experiments
4. PPI Experiments
5. Data Standards
6. Data Sources

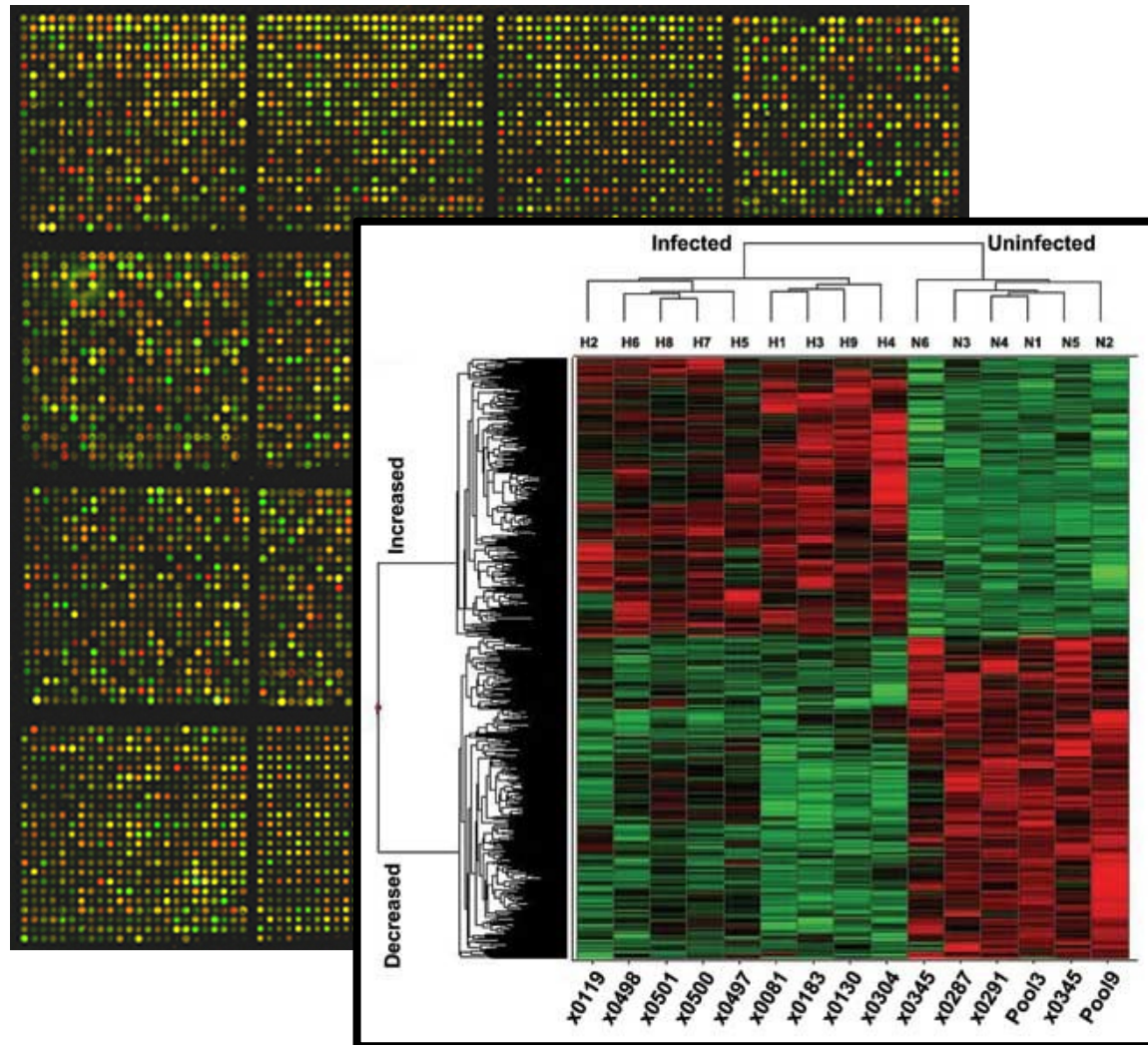
Transcriptomics

- Measure expression level of all RNAs
- Tissue or single cell
- Measure of which Genes are being expressed in a cell/tissue



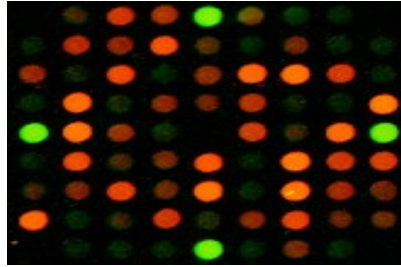
Transcriptomics

- Estimates of mRNA concentrations
- Large open databases (GEO, ArrayExpress)
- Standardised data formats, MAGE-ML, MAGE-TAB, MIAME, HUP-ML
- Reliable Open Source Software (Bioconductor)

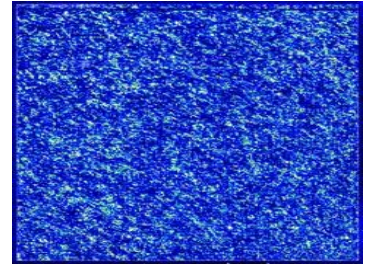


Experimental methods

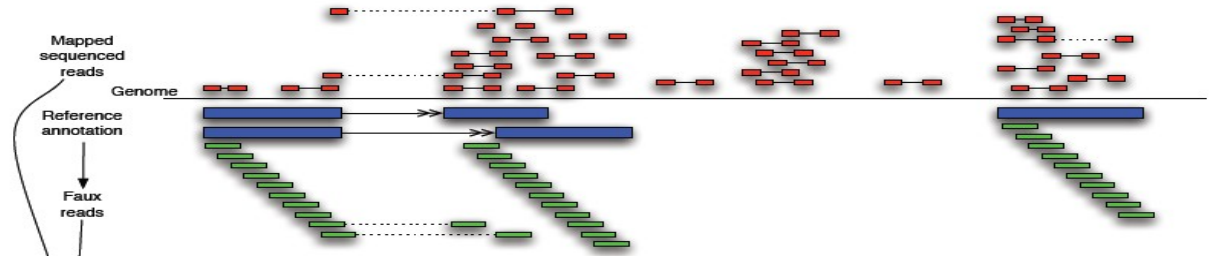
1. Spotted microarrays



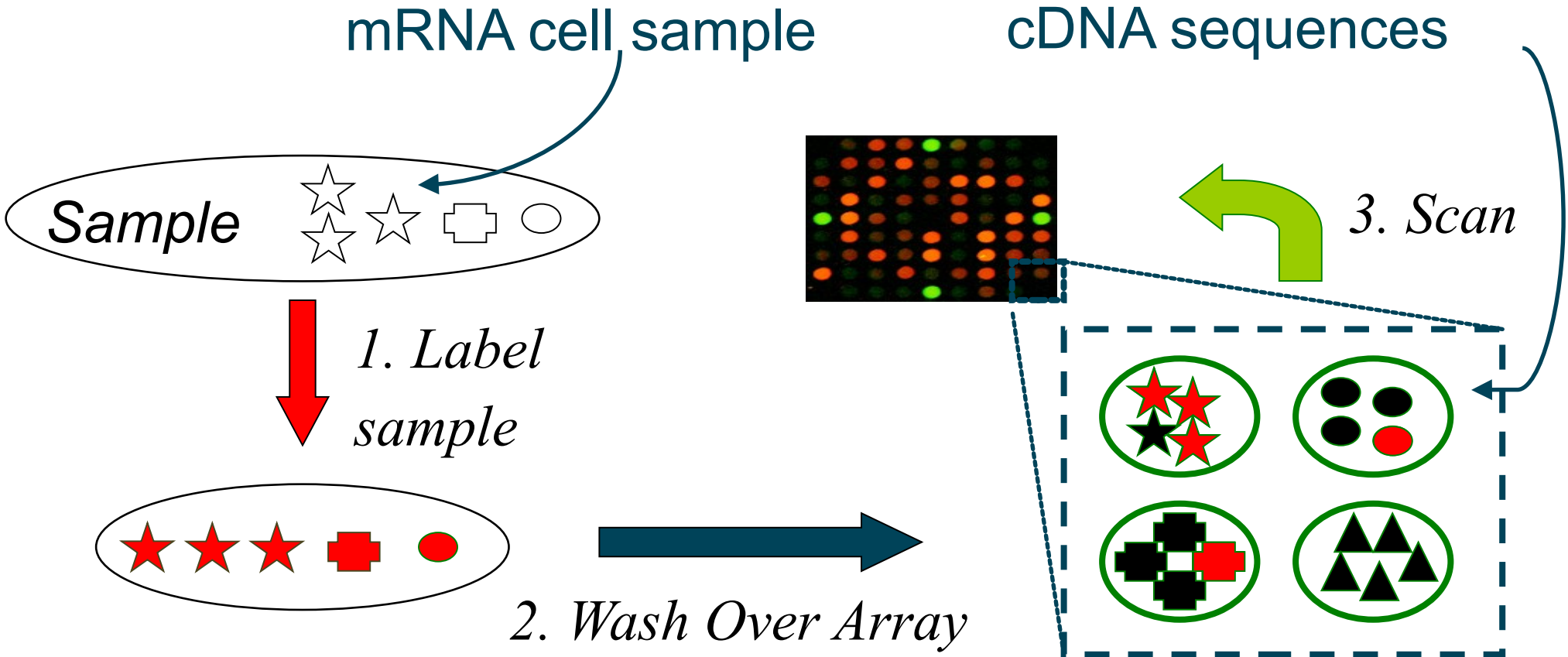
2. Affymetrix GeneChips



3. RNASeq



Microarrays overview: Each spot has “complementary” DNA attached to it for a particular gene



Spotted Microarrays – an example

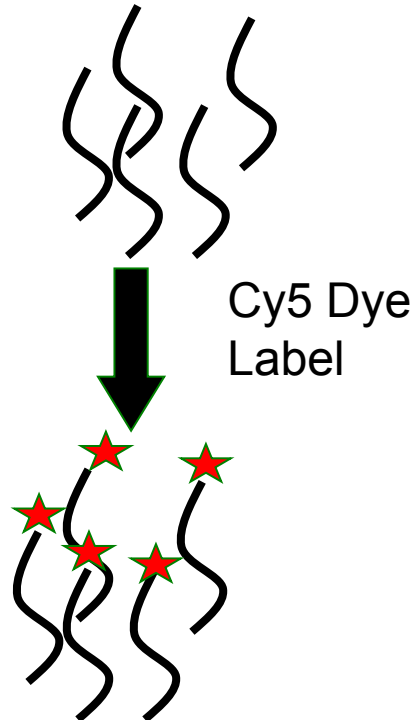
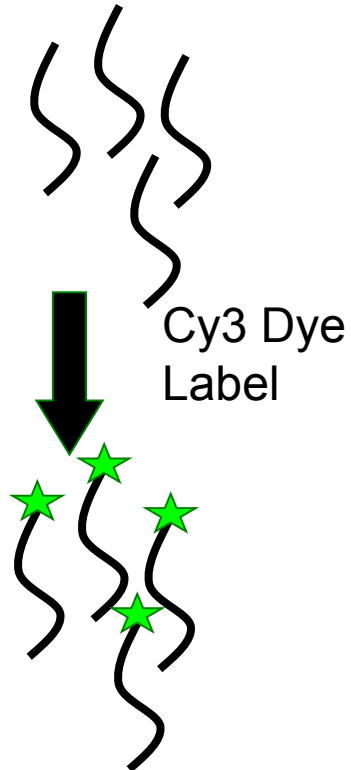
- As an example, we will look at detecting differences between acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML) . This was one of the first successful uses of microarrays in cancer classification.
- Total mRNA was extracted from bone marrow taken from patients with ALL and AML.
- Converted to cDNA by reverse transcription (since mRNA is sensitive to degradation).

T.R. Golub et. al (1999) Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring. Science, vol. 286, 531-537.

Spotted Microarray Experimental Design – Direct Comparison using 2 Channels

ALL cDNAs

AML cDNAs



Spot Interpretation

Green Spot – higher expression in ALL.

Red Spot – higher expression in AML.

Yellow Spot – equal expression in both.

Black Spot – not expressed in either.



Mix and hybridize onto slide

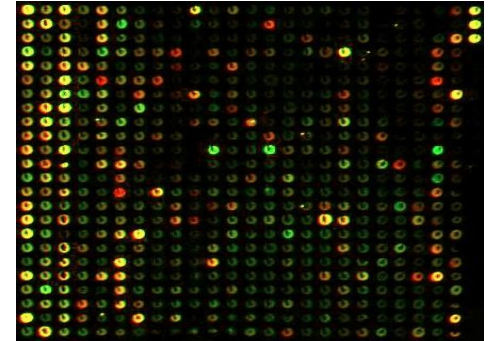
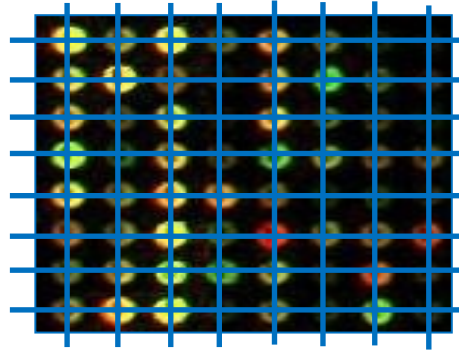
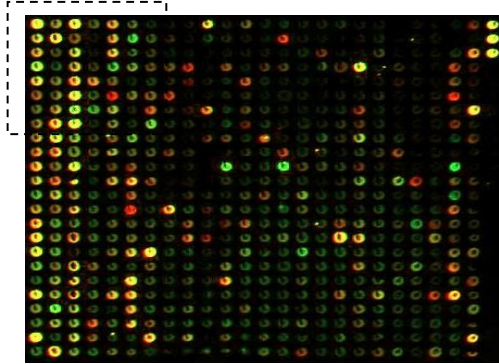
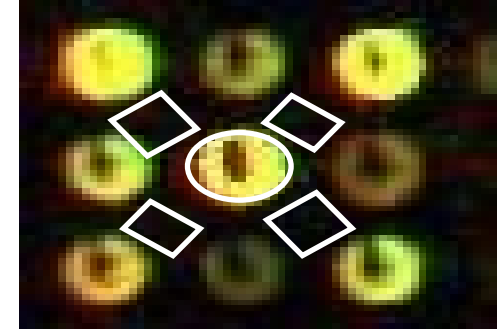


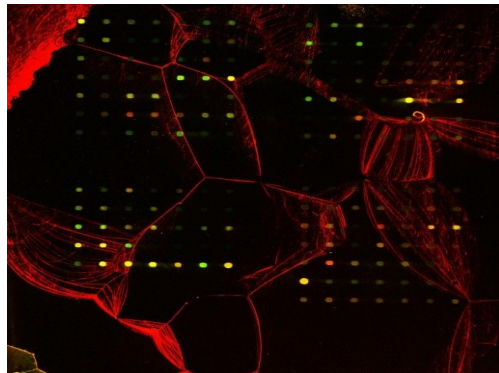
Image Analysis – Spotted Arrays



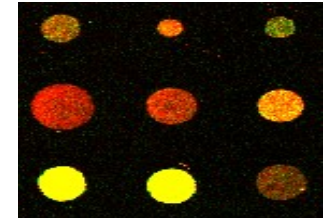
Gridding



Segmentation and
background
extraction



Complications include spots lying on curves, spots of different shapes and sizes, variation in background fluorescence, etc.



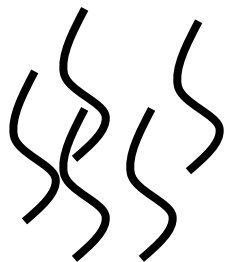
Interpreting the Spot Colour

- Using two channels helps eliminate a lot of unwanted variation (e.g. spot shapes) on the chip since these affect both channels.
- Two colour arrays produce two values, the cy5 and cy3 channels
 - these are divided to produce a ratio: $\text{cy5}/\text{cy3}$ where cy5 is red intensity (sample 1) and cy3 is green intensity (sample 2).

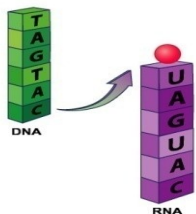
Expression values								
	Cy5	5000	10000	6000	20000	15000	10000	16000
	Cy3	40000	40000	12000	20000	7500	2500	2000
Ratio		1/8	1/4	1/2	1	2/1	4/1	8/1
Log2 Ratio		-3	-2	-1	0	1	2	3

Affymetrix GeneChip – Single Channel

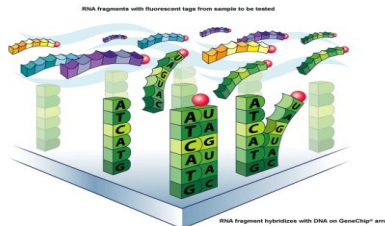
ALL cDNAs



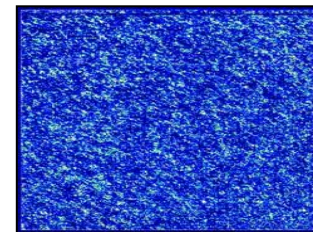
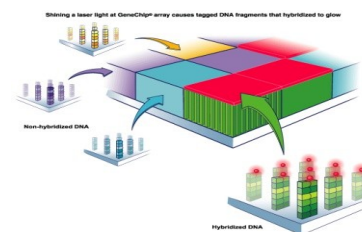
Biotin Label



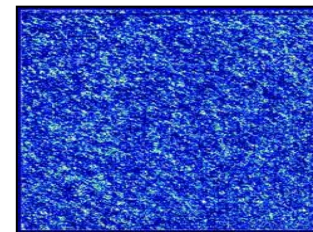
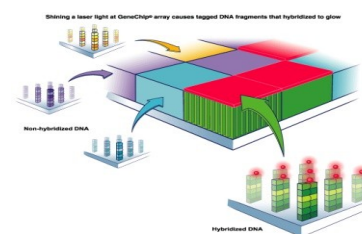
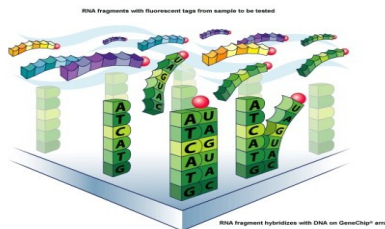
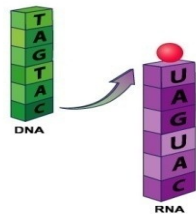
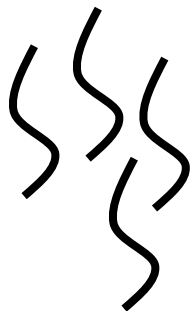
Hybridize



Scan

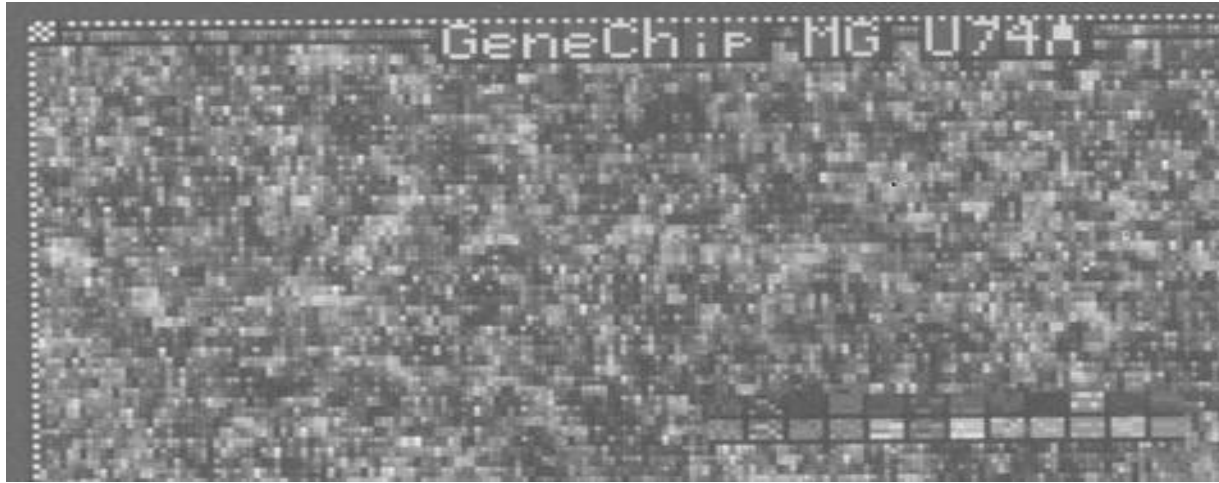


AML cDNAs

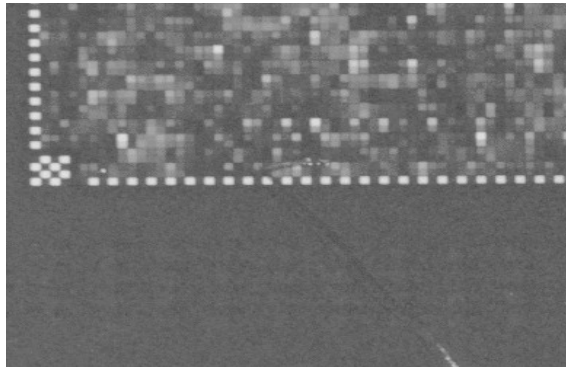


Multiple spots per gene

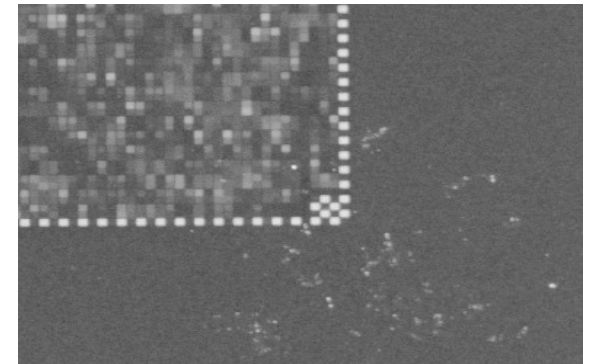
Image Analysis – Affymetrix



Affymetrix manufacturing and processing is much more controlled and thus image analysis is more straightforward (generally using their own software).

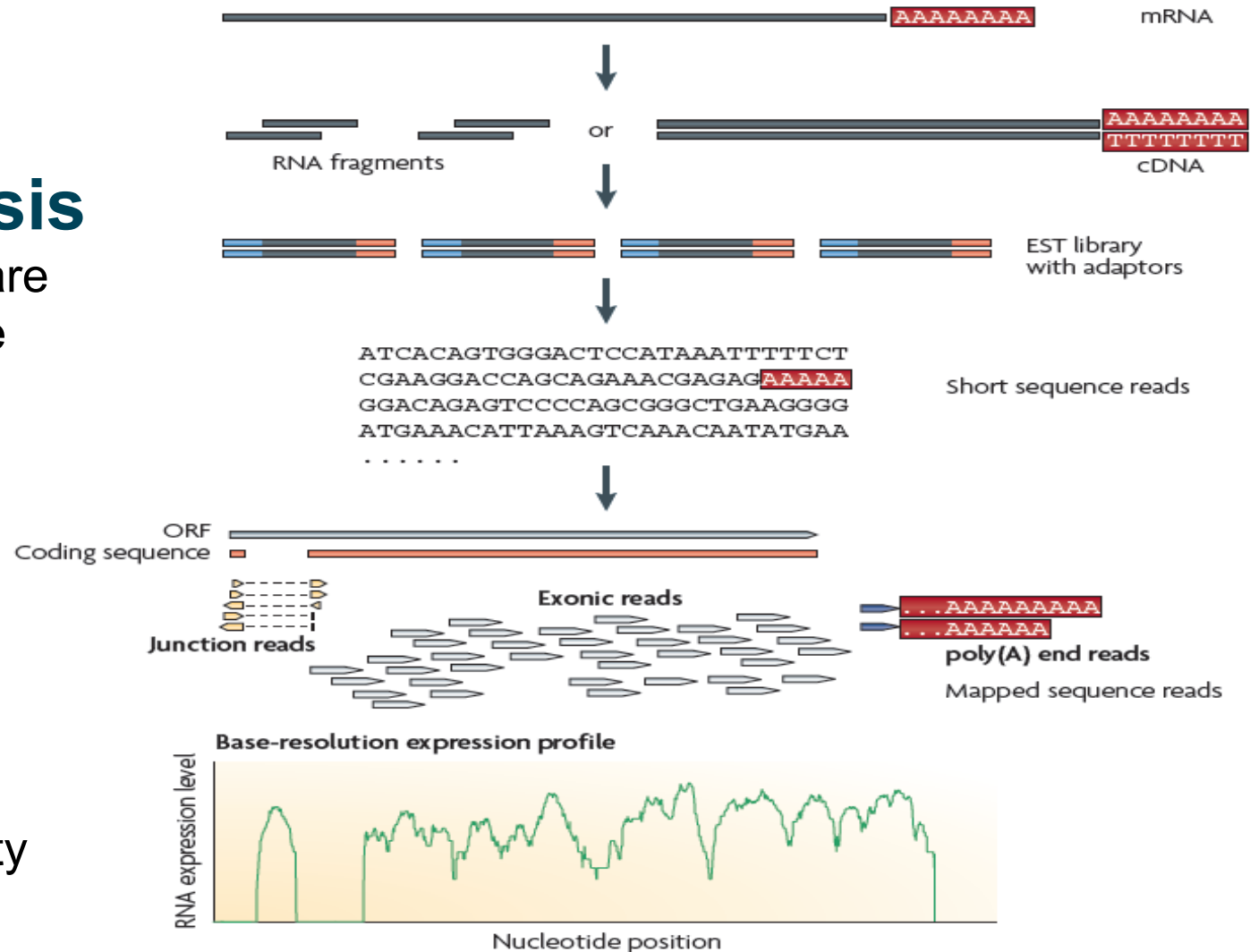


But Affymetrix can still suffer from surface defects such as scratches and visual inspection is important.



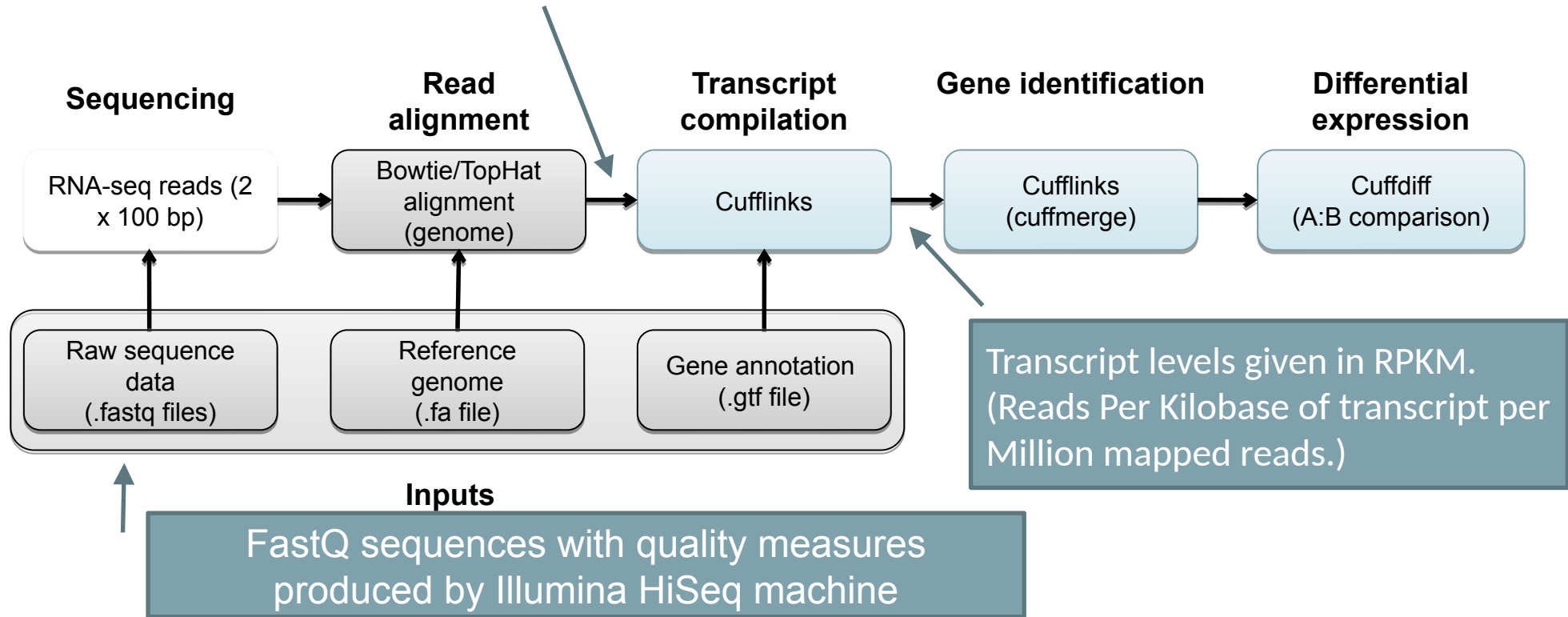
RNASeq – basic data analysis

- Essentially short reads are mapped onto a reference genome to get counts of gene expression.
- Not limited to known sequences (as for hybridization techniques)
- Very low, if any, background and capability to detect very low expressed genes

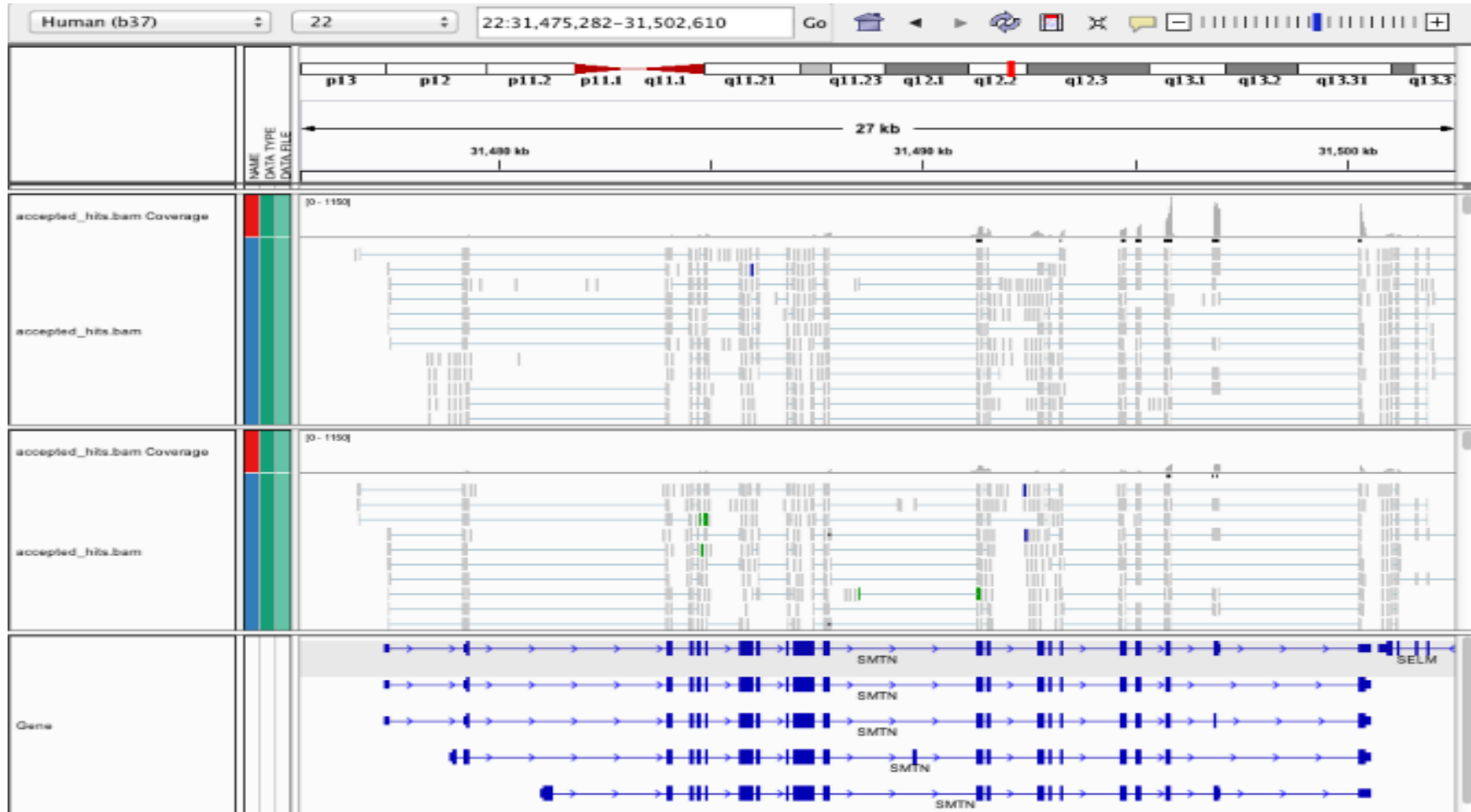


Bowtie/Tophat/Cufflinks/Cuffdiff RNA-seq Pipeline

SAM Files = Sequence Alignment Map Format
BAM Files = Binary compressed SAM Files

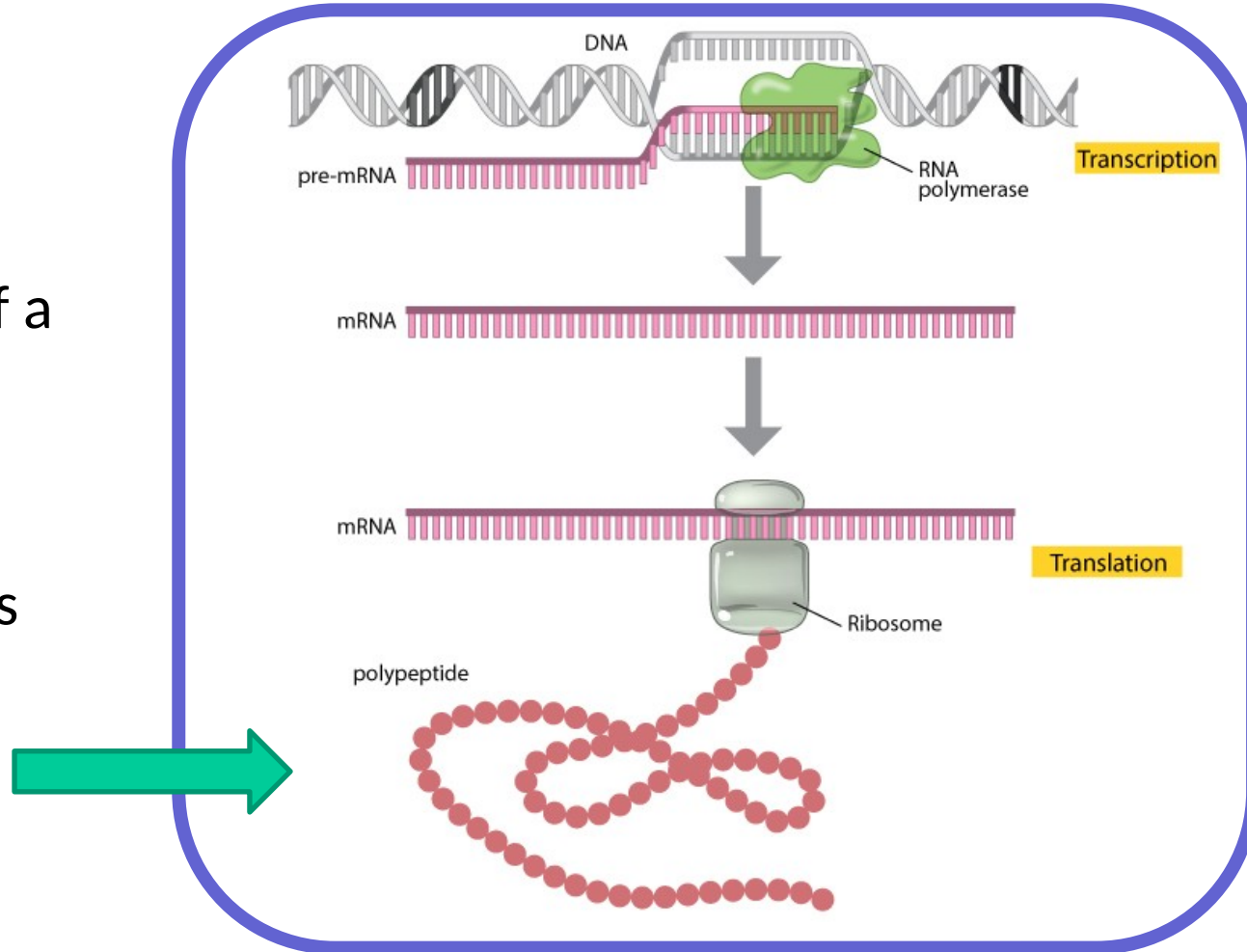


IGV (Integrated Genome Viewer) showing reads



Proteomics

- Measure the concentration/presence of a protein in a tissue
- Direct estimate of what protein activities/functions are present in a tissue



Biological replicates are important ...

In the ALL and AML study we have 27 independent samples from patients with ALL and 11 independent samples from patients with AML

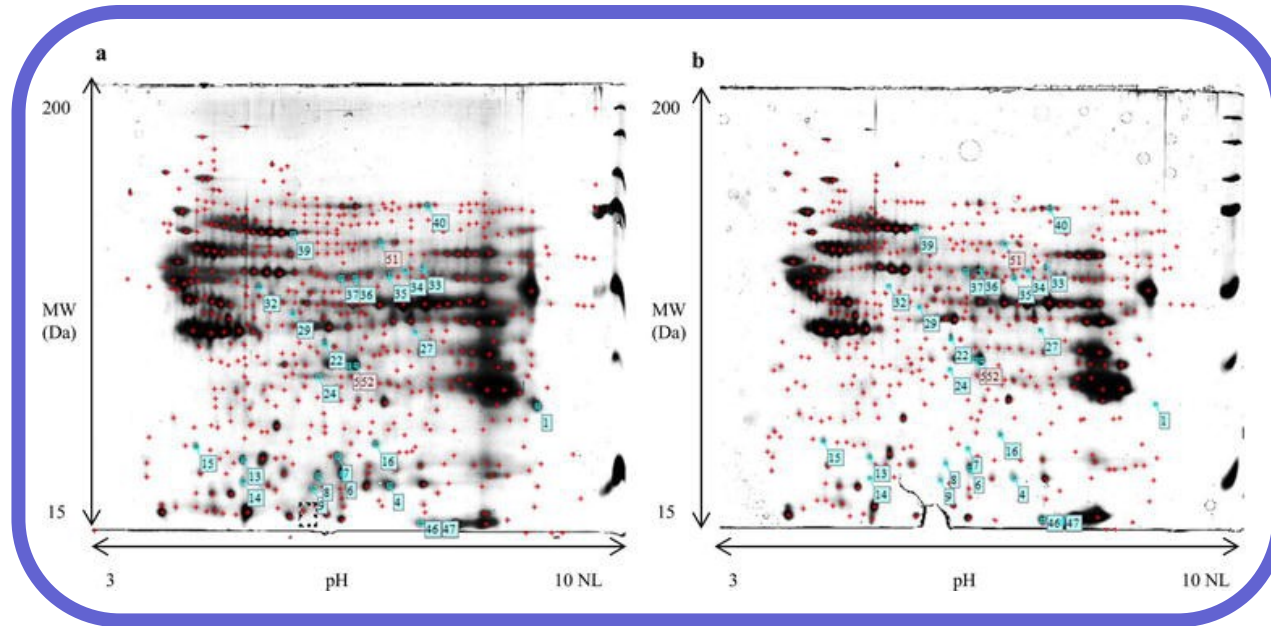
These are **biological replicates**, in that they provide information about the variation between biological samples.

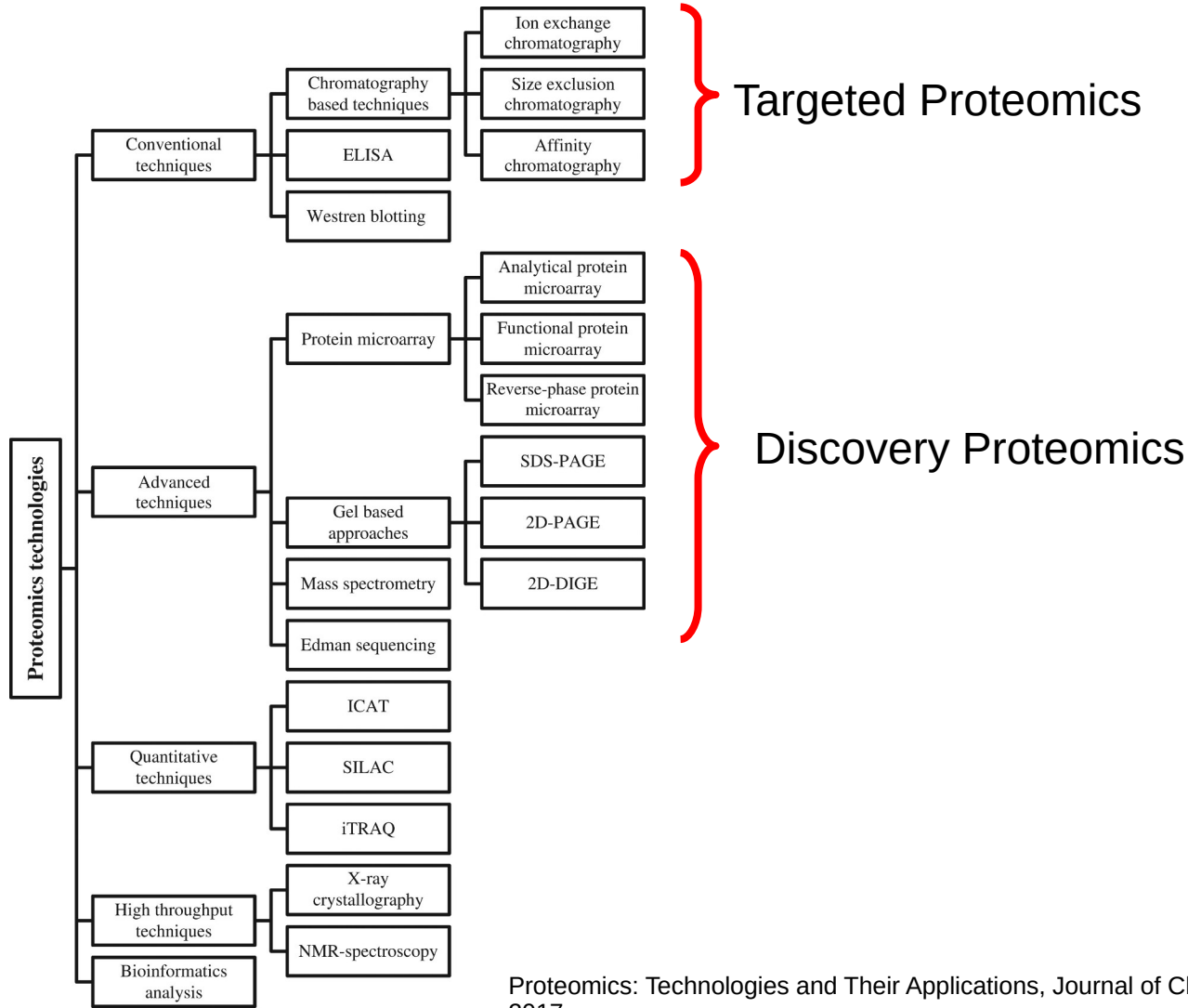
High-throughput data is noisy: You should aim to have at least 5 biological replicates (an absolute minimum of 3 !).

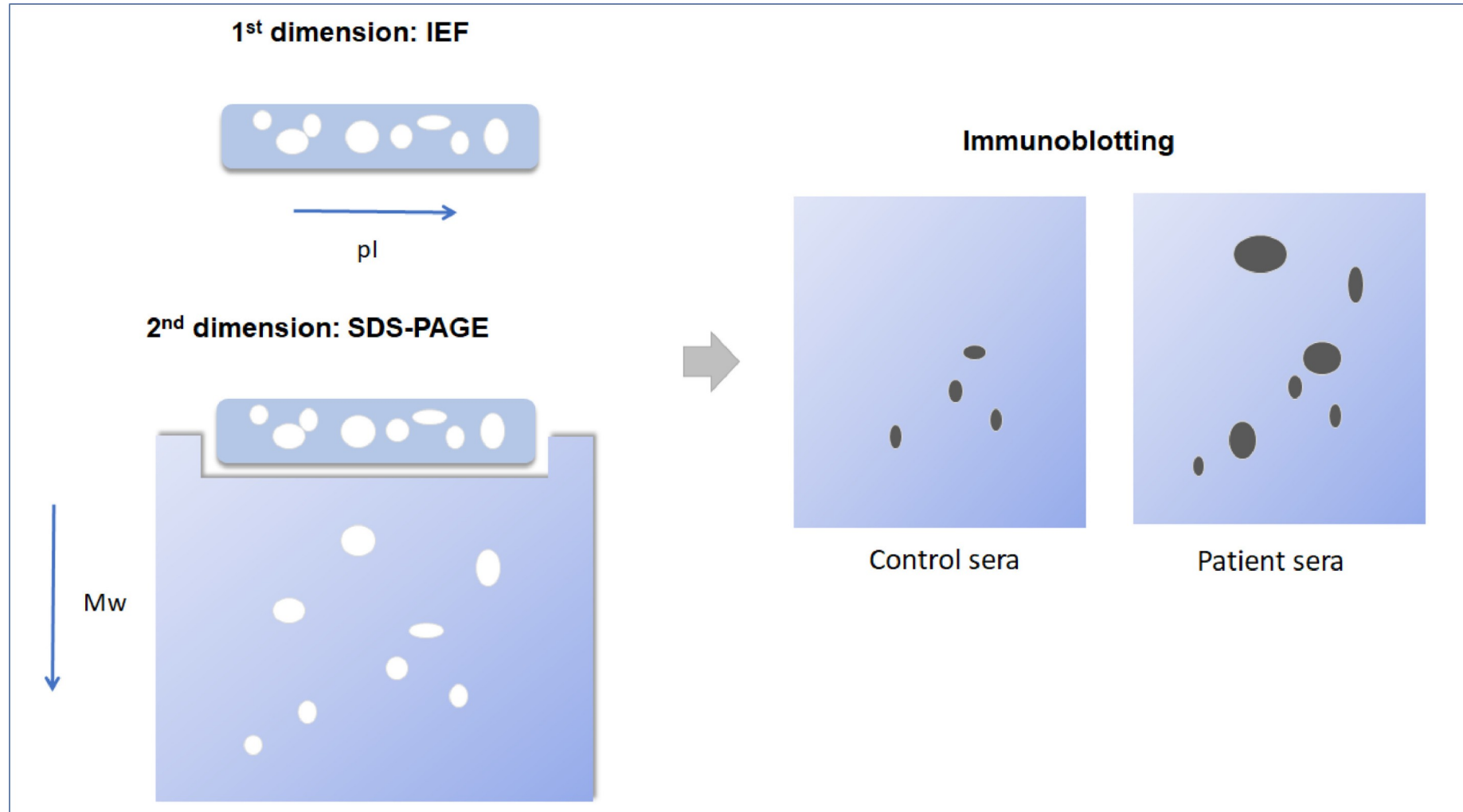
Replicates are essential in transcriptomics studies, they not only make the mean expression values more accurate (reducing random noise), but also provide information about the variability of a particular expression value in the natural biological population.

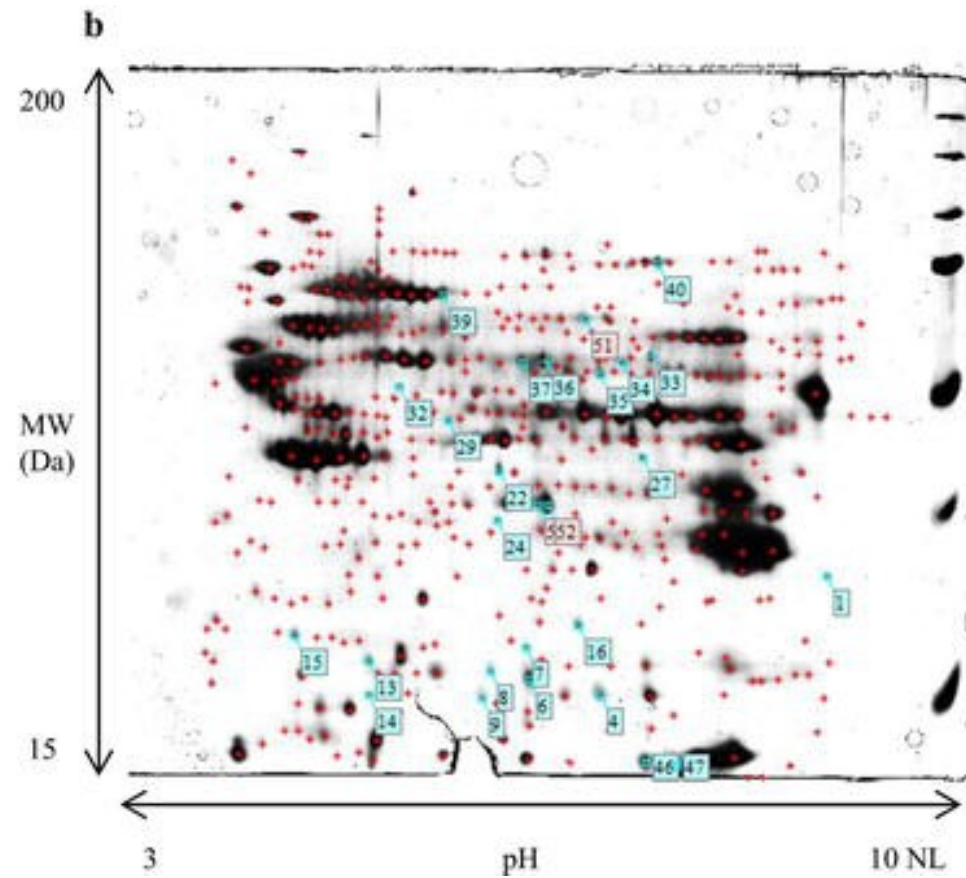
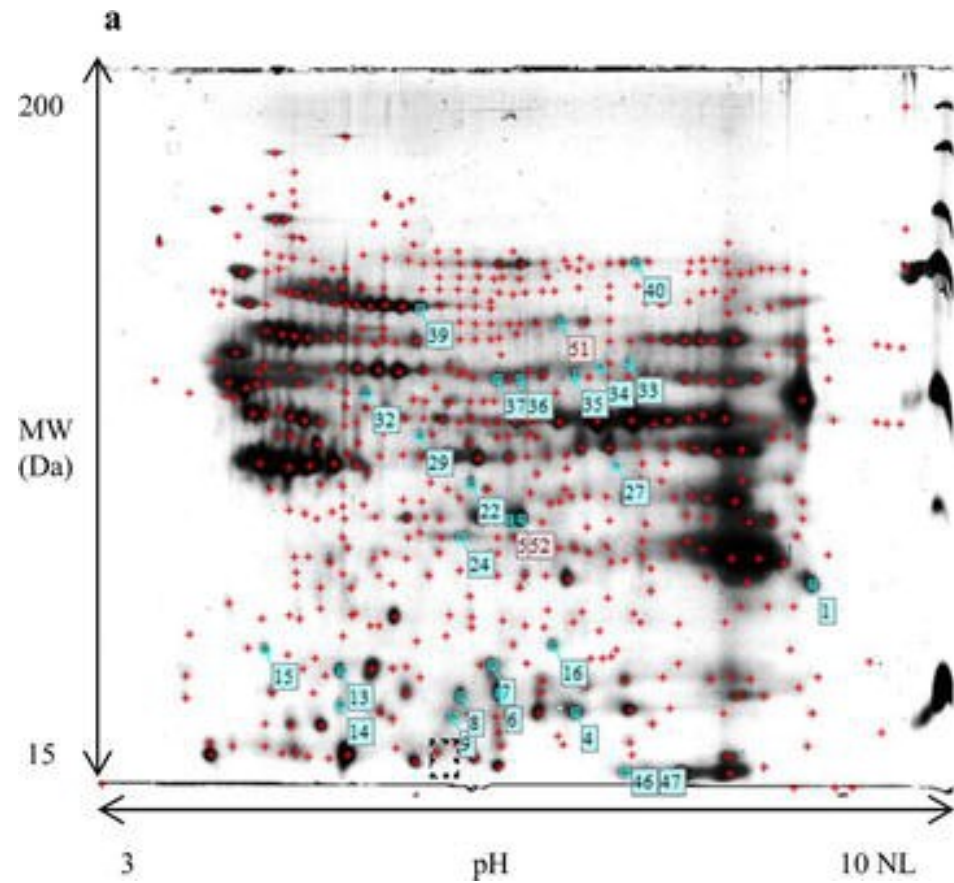
Proteomics

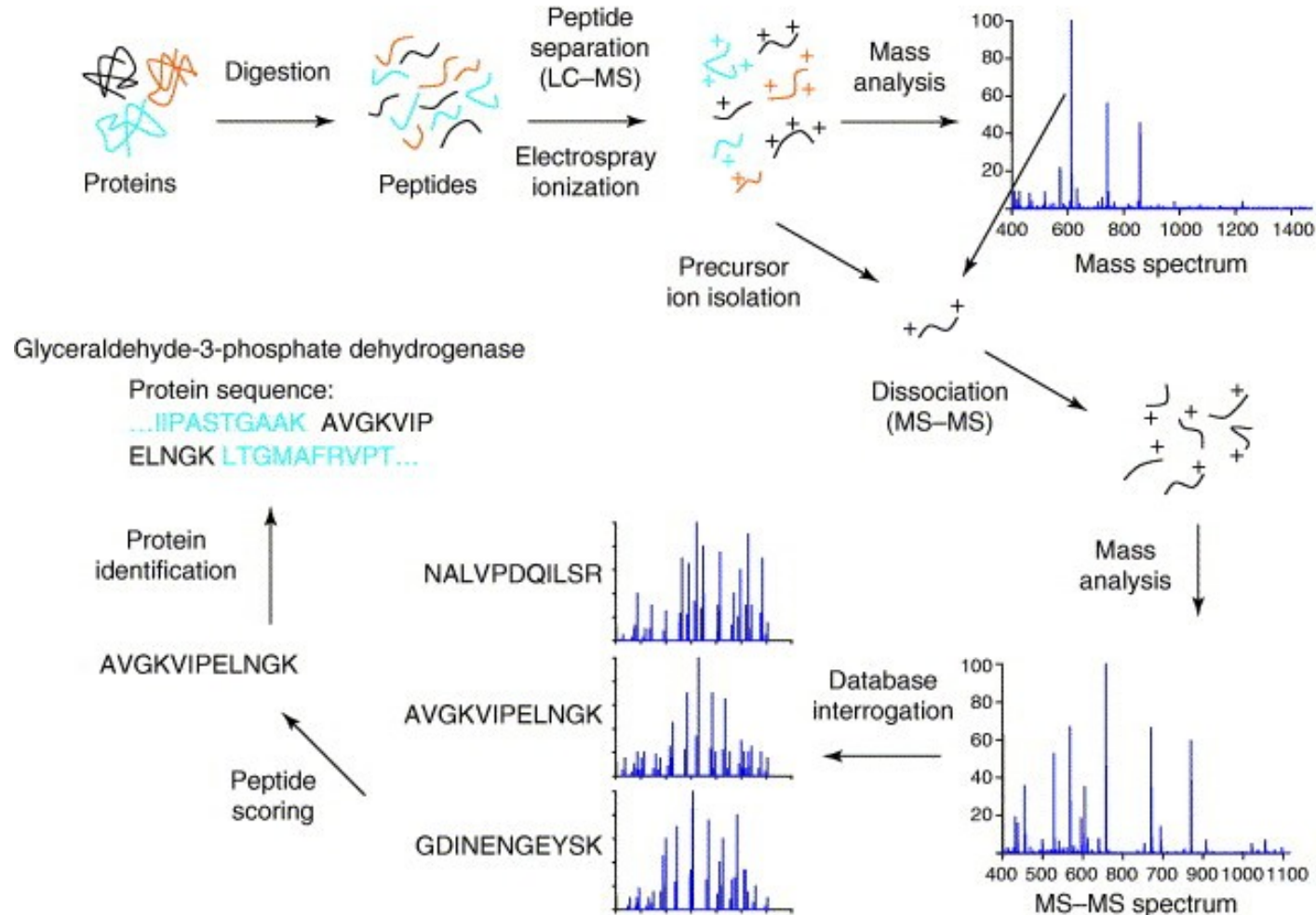
- Measure the concentration/presence of a protein in a tissue
- Direct estimate of what protein activities/functions are present in a tissue
- Experimental systems: 2D PAGE, Mass Spec', Protein Chips.
- Software/data standards
<https://psidev.info/>





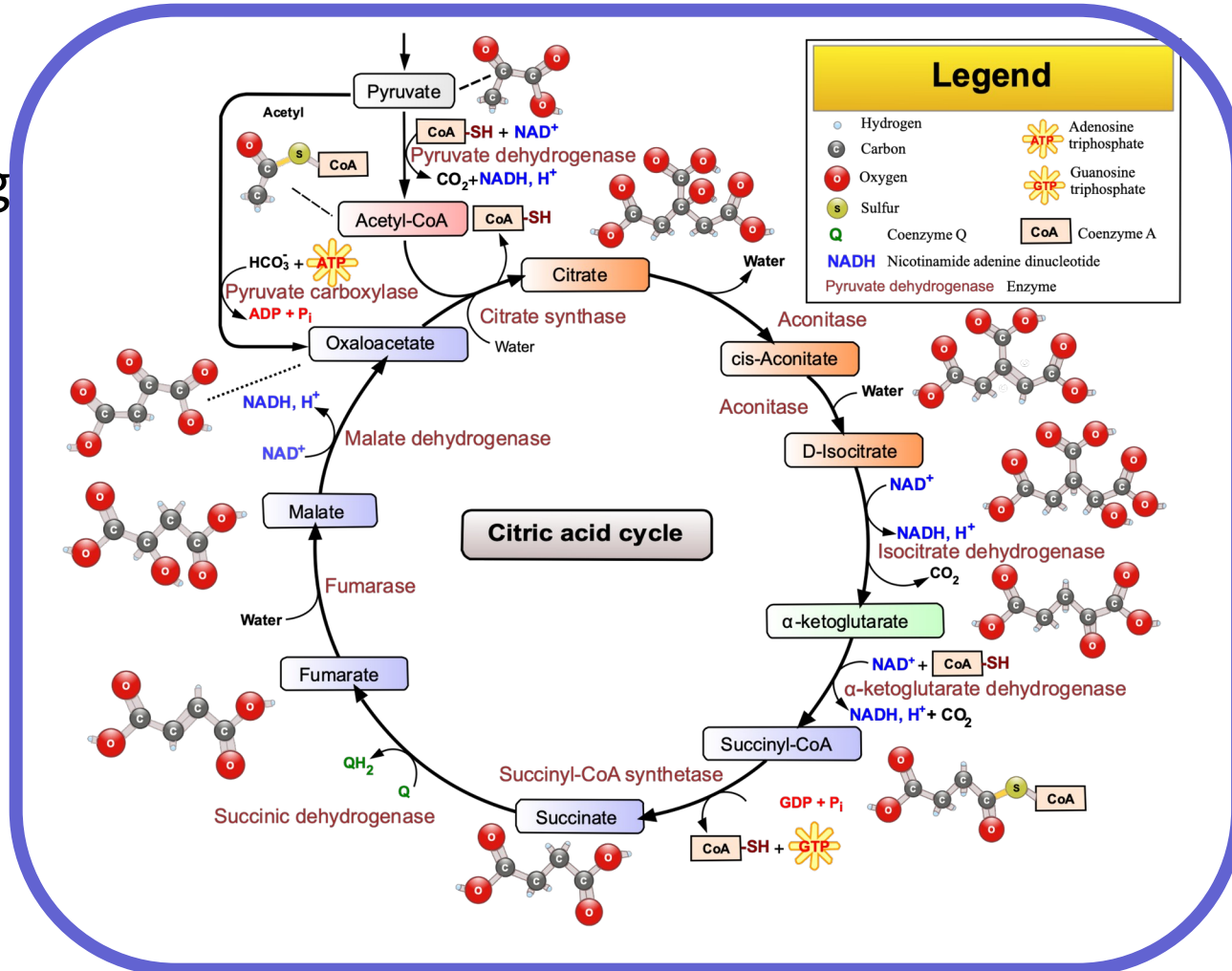






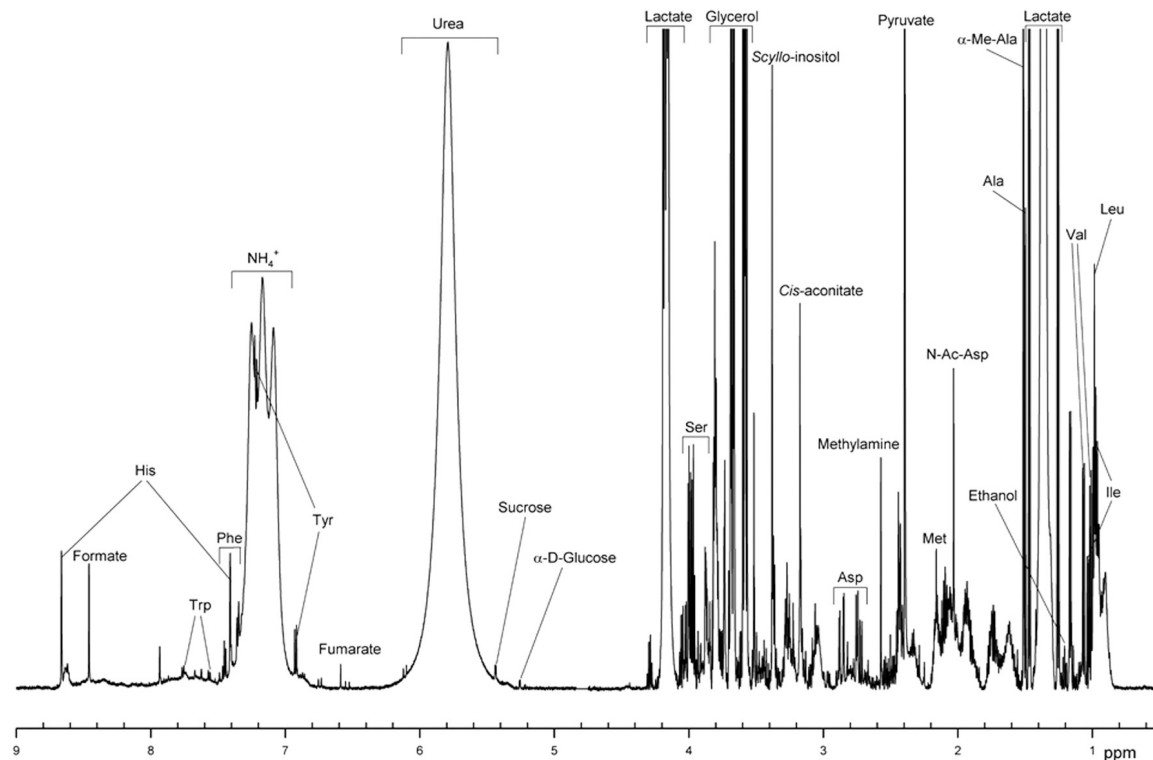
Metabolomics

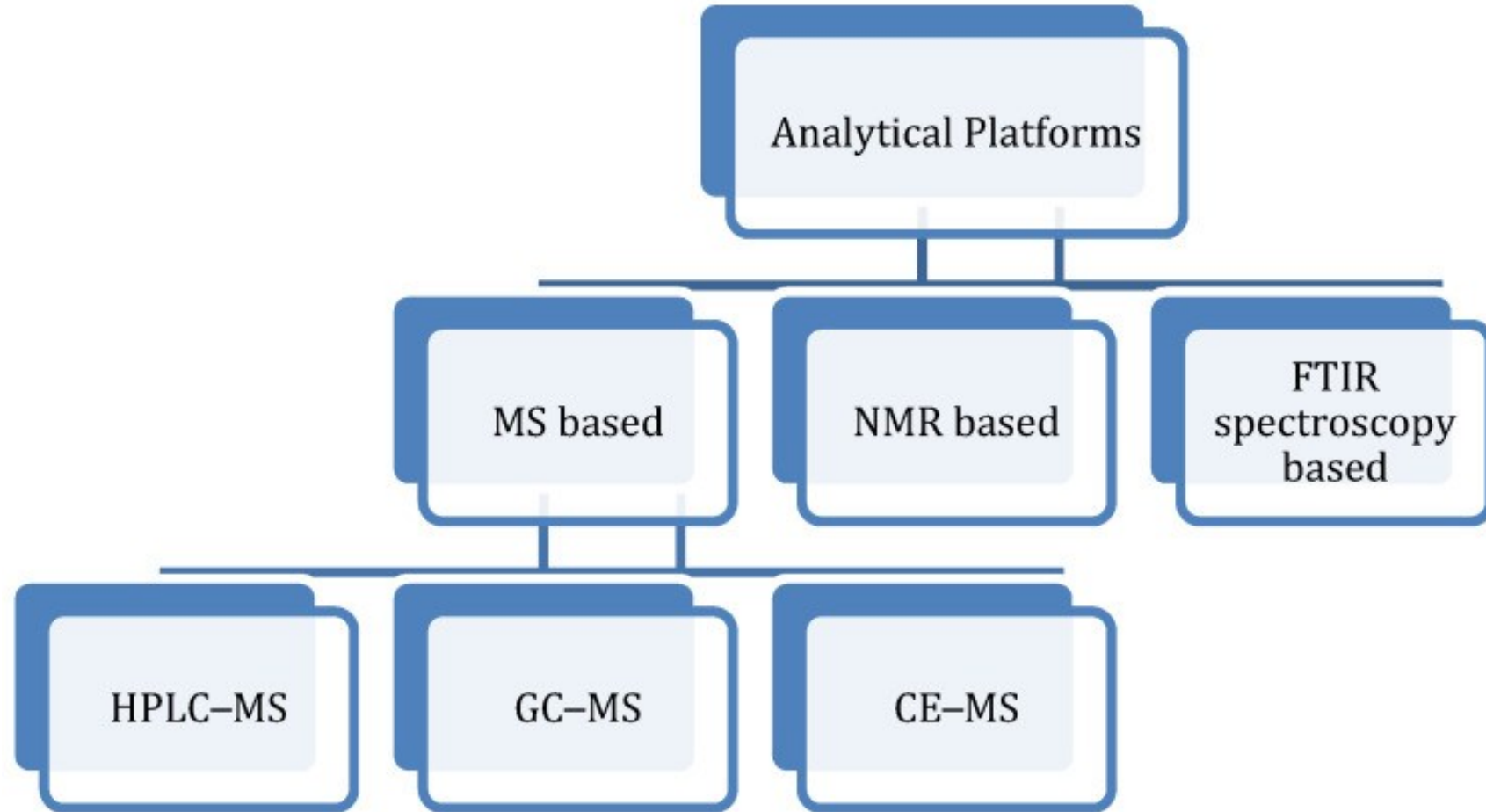
- Cataloguing and measuring the presence and concentration of all metabolites in a tissue

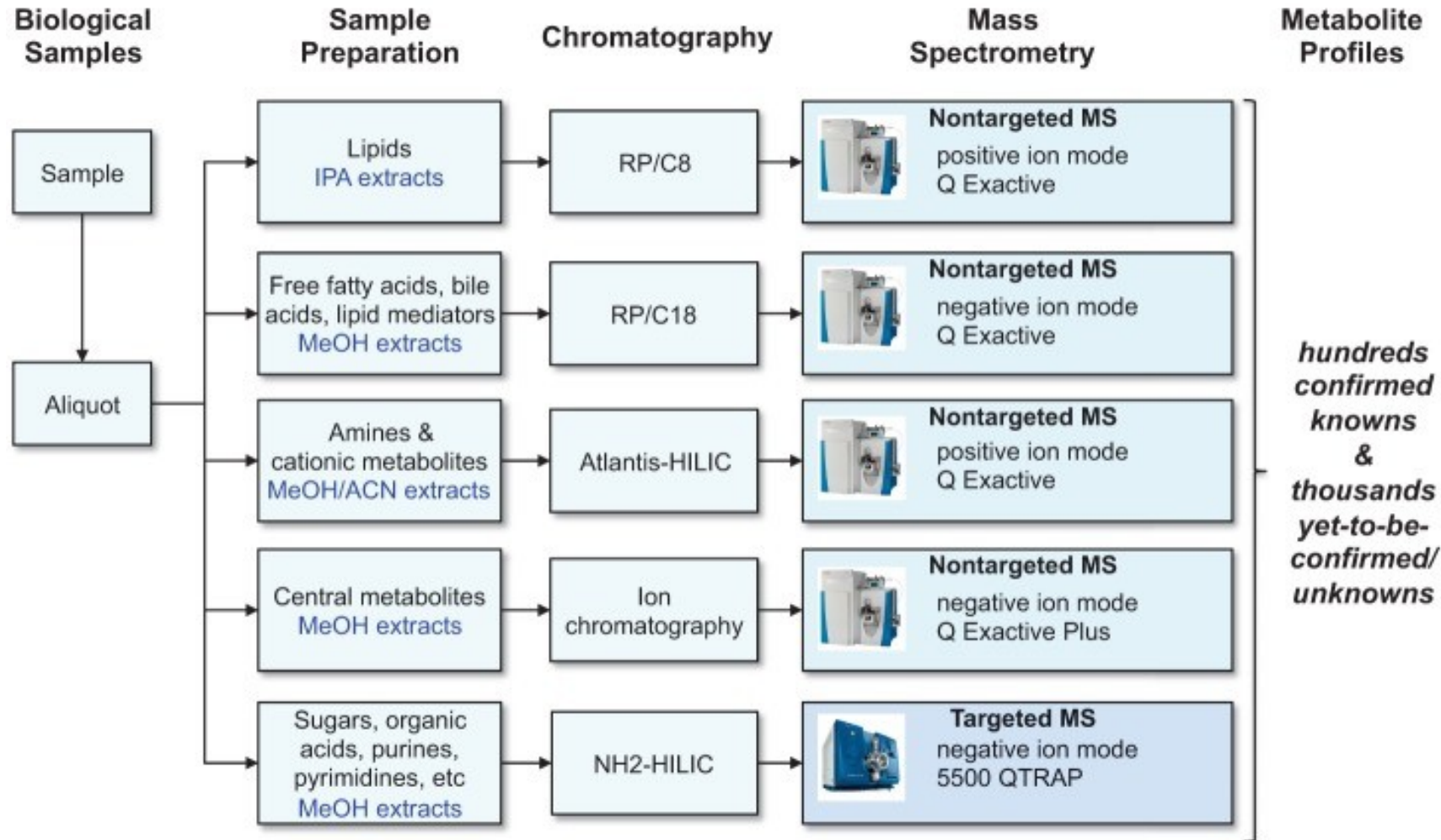


Metabolomics

- Small molecule catalogues under differing conditions
- High resolution NMR data
- Data standards in development
- <https://msi-workgroups.sourceforge.net>

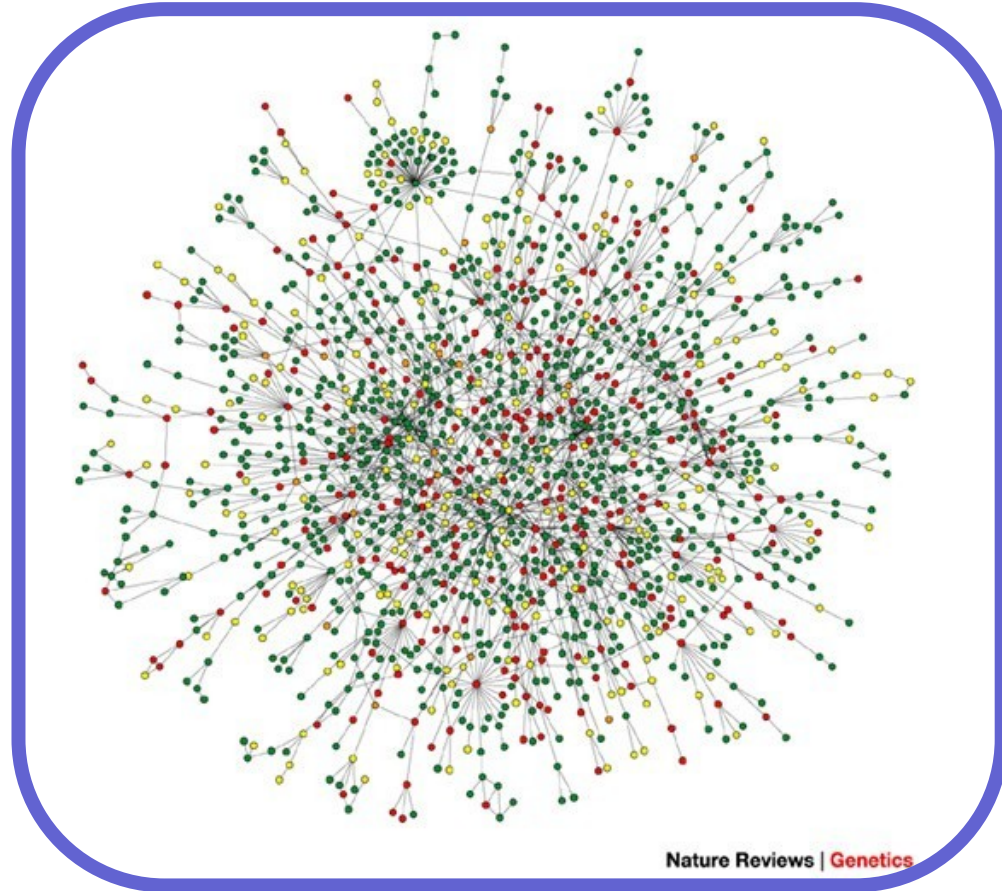






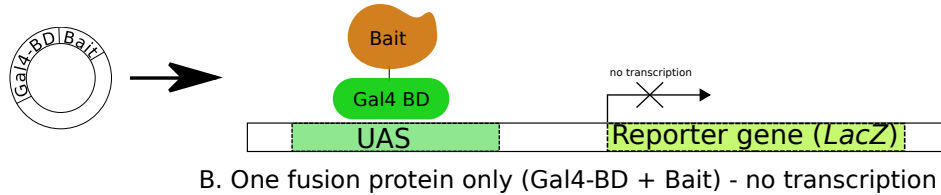
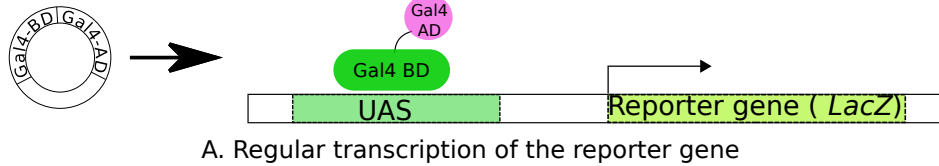
Protein-Protein Interaction Networks

- Protein interactions under given experimental conditions
- Canonical/Ur experiment: Yeast Two-Hybrid
- Software/data standards, These guys again!
<https://psidev.info/>



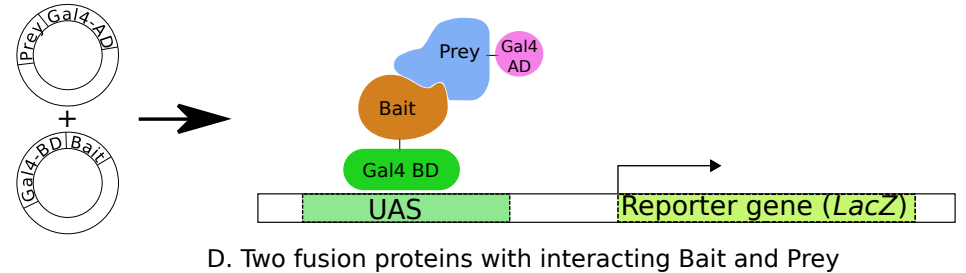
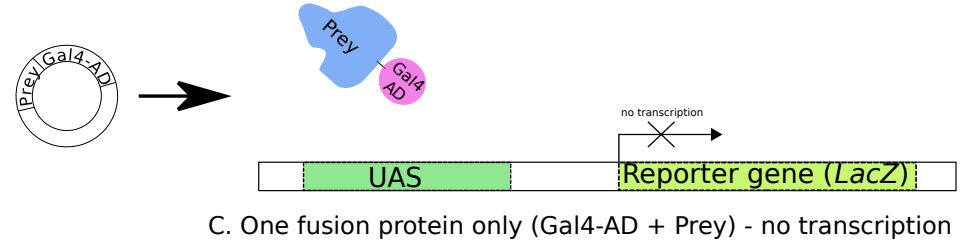
Yeast Two Hybrid Screening

A) Wild Type



B) TF with Bait Protein

C) TF with Bait Protein



D) Binding Re-establishes TF ability

Data Standards and Data Sources

How information about microarray experiments was often communicated

Had to read the research paper and digest the structure of the experiment ...

REPORTS

Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring

T. R. Golub,^{1,2,*†} D. K. Slonim,^{1†} P. Tamayo,¹ C. Huard,¹ M. Gaasenbeek,¹ J. P. Mesirov,¹ H. Coller,¹ M. L. Loh,² J. R. Downing,³ M. A. Caligiuri,⁴ C. D. Bloomfield,⁴ E. S. Lander^{1,5*}

Although cancer classification has improved over the past 30 years, there has been no general approach for identifying new cancer classes (class discovery) or for assigning tumors to known classes (class prediction). Here, a generic approach to cancer classification based on gene expression monitoring by DNA microarrays is described and applied to human acute leukemias as a test case. A class discovery procedure automatically discovered the distinction between acute myeloid leukemia (AML) and acute lymphoblastic leukemia (ALL) without previous knowledge of these classes. An automatically derived class predictor was able to determine the class of new leukemia cases. The results demonstrate the feasibility of cancer classification based solely on gene expression monitoring and suggest a general strategy for discovering and predicting cancer classes for other types of cancer, independent of previous biological knowledge.

The challenge of cancer treatment has been to target specific therapies to pathogenetically distinct tumor types, to maximize efficacy and minimize toxicity. Improvements in cancer classification have thus been central to advances in cancer treatment. Cancer classification has been based primarily on morphological appearance of the tumor, but this has serious limitations. Tumors with similar histopathological appearance can follow significantly different clinical courses and show different responses to therapy. In a few cases, such clinical heterogeneity has been explained by dividing morphologically similar tumors into subtypes with distinct pathogenesis. Key examples include the subdivision of acute leukemias, non-Hodgkin's lymphomas, and childhood "small round blue cell tumors" [tumors with variable response to chemotherapy (1) that are now molecularly subclassified into neuroblastomas, rhabdomyosarcoma, Ewing's sarcoma, and other types (2)]. For many more tumors, important subclasses are likely to exist but have yet to

be defined by molecular markers. For example, prostate cancers of identical grade can have widely variable clinical courses, from indolence over decades to explosive growth causing rapid patient death. Cancer classification has been difficult in part because it has historically relied on specific biological insights, rather than systematic and unbiased approaches for recognizing tumor subtypes. Here we describe such an approach based on global gene expression analysis.

We divided cancer classification into two challenges: class discovery and class prediction. Class discovery refers to defining previously unrecognized tumor subtypes. Class prediction refers to the assignment of particular tumor samples to already-defined classes, which could reflect current states or future outcomes.

We chose acute leukemias as a test case. Classification of acute leukemias began with the observation of variability in clinical outcome (3) and subtle differences in nuclear morphology (4). Enzyme-based histochemical analyses were introduced in the 1960s to demonstrate that some leukemias were peroxidase positive, whereas others were myeloperoxidase positive (5). This provided the first basis for classification of acute leukemias into those arising from lymphoid precursors (acute lymphoblastic leukemia, ALL) or from myeloid precursors (acute myeloid leukemia, AML). This classification was further solidified by the development in the 1970s of antibodies recognizing either lymphoid or myeloid cell surface molecules (6). Most recently, particular subtypes of

acute leukemia have been found to be associated with specific chromosomal translocations—for example, the t(12;21)(p13;q22) translocation occurs in 25% of patients with ALL, whereas the t(8;21)(q22;q22) occurs in 15% of patients with AML (7).

Although the distinction between AML and ALL has been well established, no single test is currently sufficient to establish the diagnosis. Rather, current clinical practice involves an experienced hematopathologist's interpretation of the tumor's morphology, histochemistry, immunophenotyping, and cytogenetic analysis, each performed in a separate, highly specialized laboratory. Although usually accurate, leukemia classification remains imperfect and errors do occur.

Distinguishing ALL from AML is critical for successful treatment: chemotherapy regimens for ALL generally contain corticosteroids, vincristine, methotrexate, and L-asparaginase, whereas most AML regimens rely on a backbone of daunorubicin and cytarabine (8). Although remissions can be achieved using ALL therapy for AML (and vice versa), cure rates are markedly diminished, and unwarranted toxicities are encountered.

We set out to develop a more systematic approach to cancer classification based on the simultaneous expression monitoring of thousands of genes using DNA microarrays (9). It has been suggested (10) that such microarrays could provide a tool for cancer classification. Microarray studies to date (11), however, have primarily been descriptive rather than analytical and have focused on cell culture rather than primary patient material, in which genetic noise might obscure an underlying reproducible expression pattern.

We began with class prediction: How could one use an initial collection of samples belonging to known classes (such as AML and ALL) to create a "class predictor" to classify new, unknown samples? We developed an analytical method and first tested it on distinctions that are easily made at the morphological level, such as distinguishing normal kidney from renal cell carcinoma (12). We then turned to the more challenging problem of distinguishing acute leukemias, whose appearance is highly similar.

Our initial leukemia data set consisted of 38 bone marrow samples (27 ALL, 11 AML) obtained from acute leukemia patients at the time of diagnosis (13). RNA prepared from bone marrow mononuclear cells was hybridized to high-density oligonucleotide microarrays, produced by Affymetrix and containing probes for 6817 human genes (14). For each gene, we obtained a quantitative expression level. Samples were subjected to a priori quality control standards regarding the amount of labeled RNA and the quality of the scanned microarray image (15).

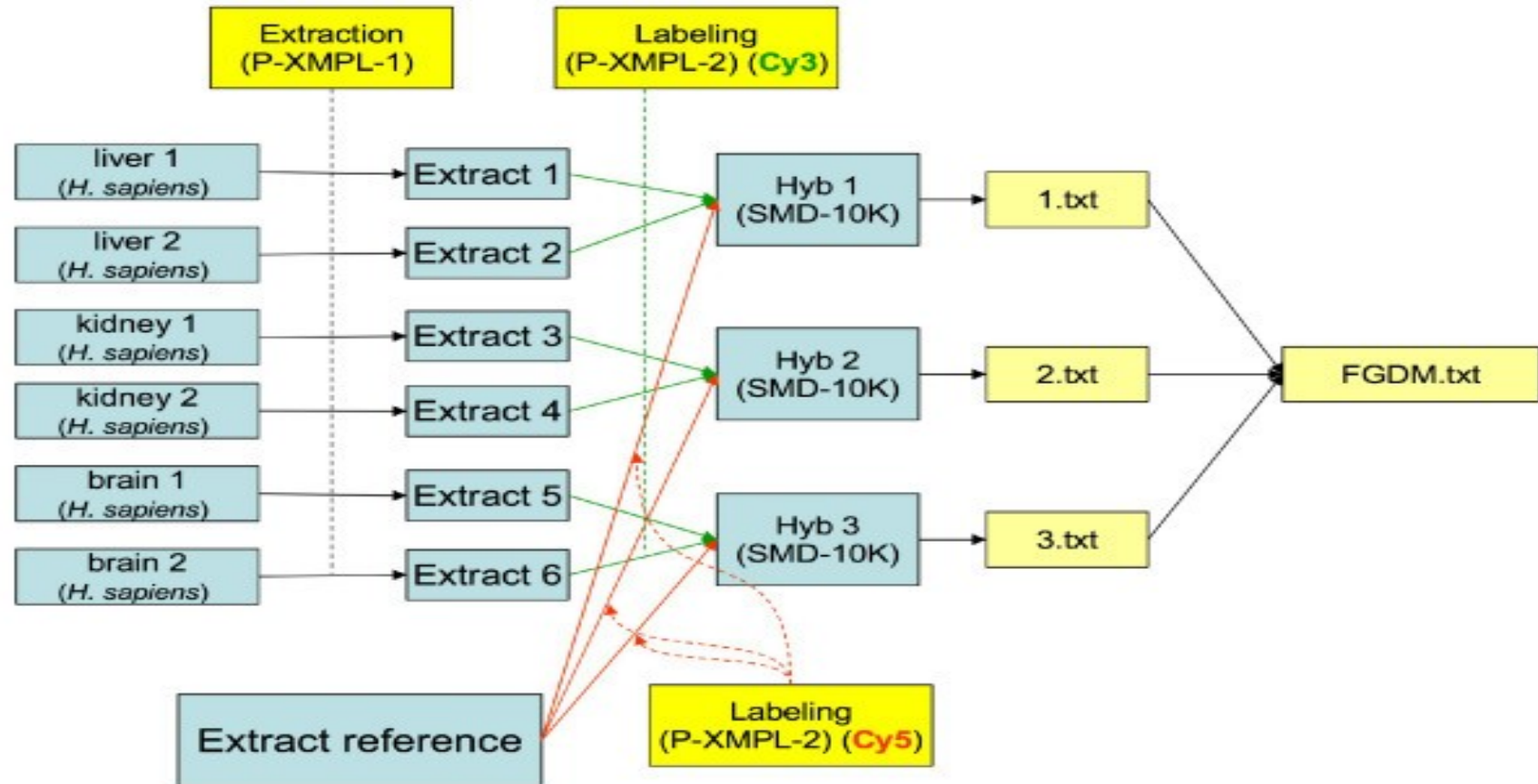
The first issue was to explore whether

¹Whitehead Institute/Massachusetts Institute of Technology Center for Genome Research, Cambridge, MA 02139, USA. ²Dana-Farber Cancer Institute and Harvard Medical School, Boston, MA 02115, USA. ³St. Jude Children's Research Hospital, Memphis, TN 38105, USA. ⁴Comprehensive Cancer Center and Cancer and Leukemia Group B, Ohio State University, Columbus, OH 43210, USA. ⁵Department of Biology, Massachusetts Institute of Technology, Cambridge, MA 02142, USA.

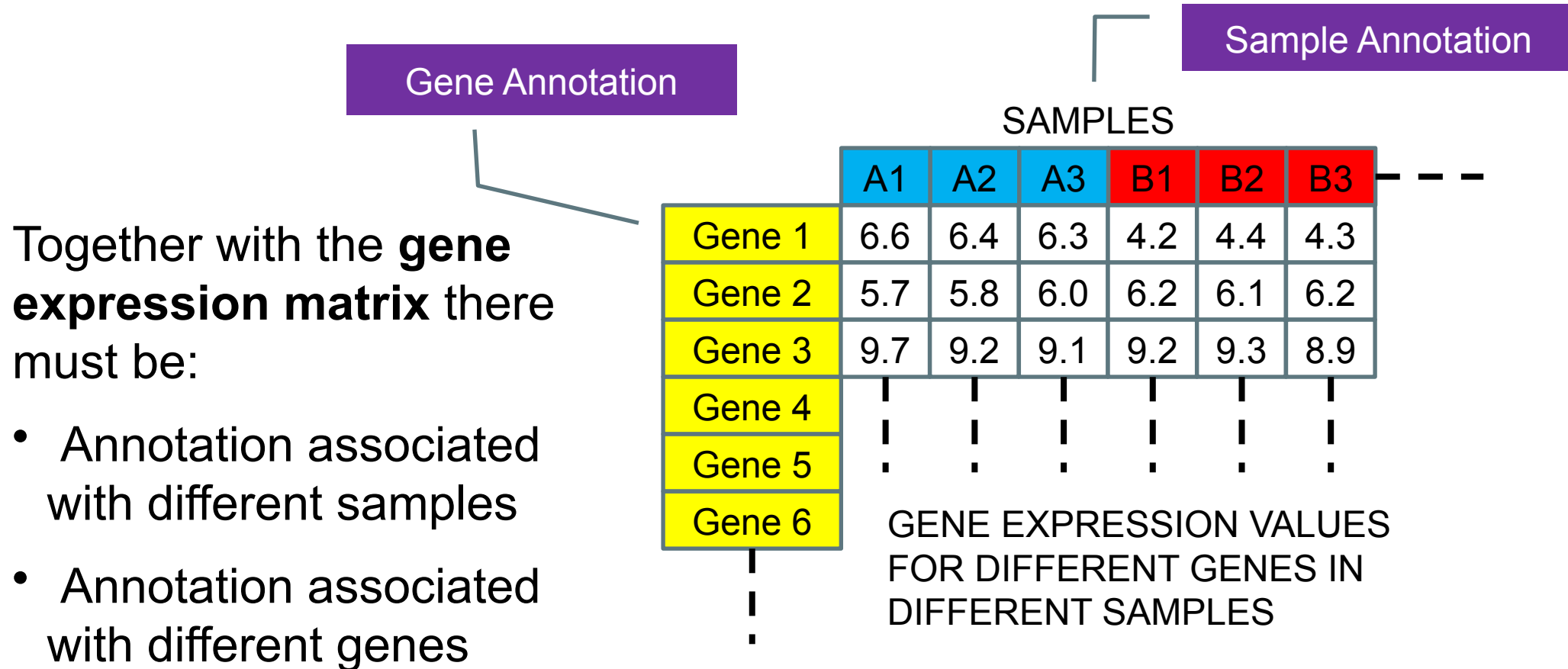
*To whom correspondence should be addressed. E-mail: golub@genome.wi.mit.edu; lander@genome.wi.mit.edu.

†These authors contributed equally to this work.

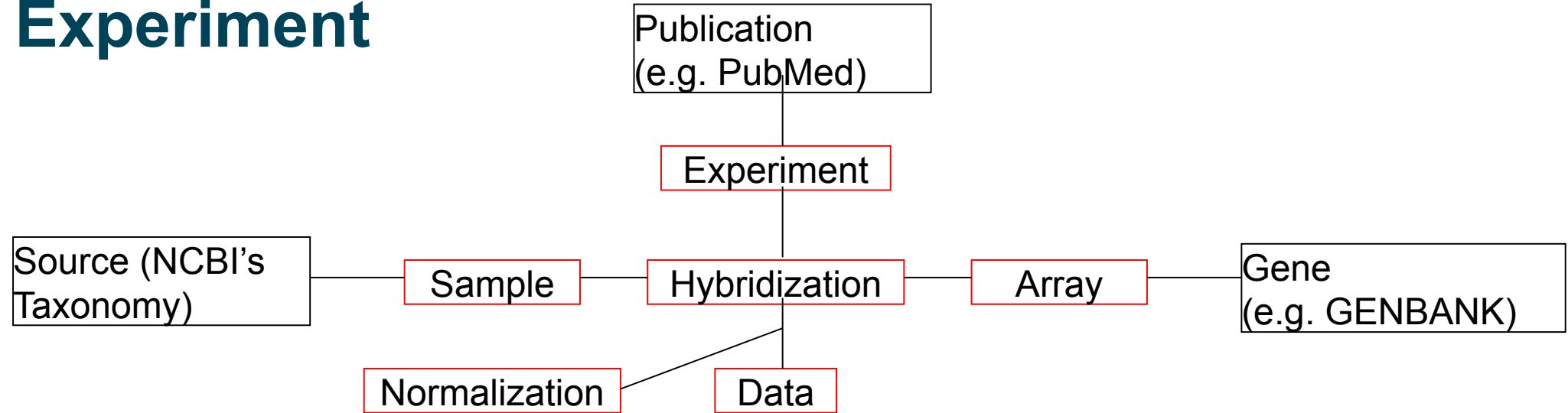
Experimental design can be complex



Transcriptomics. What do we need to capture?



MIAME: Minimum Information About a Microarray Experiment



Controlled vocabulary takes on a 'qualifier, value, source' triplet.

Qualifier: cell type

Value: epithelial

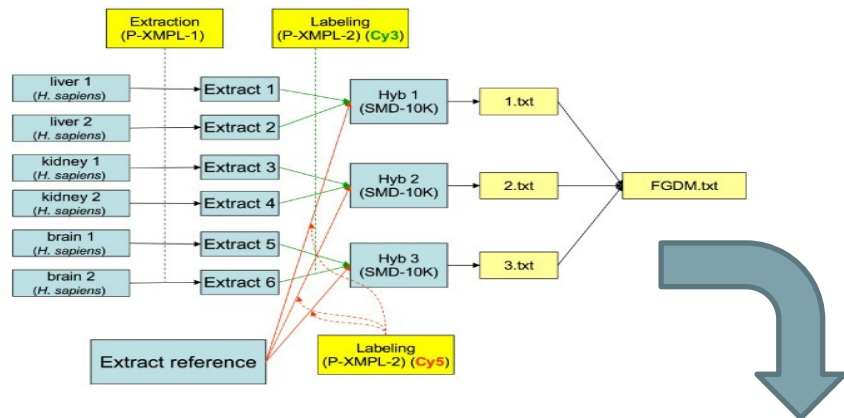
Source: Gray's anatomy (38th Ed)

Qualifier: treatment

Value: 15 min heat shock

Source: Smith and Jones, Nature Genetics (1992)

MAGE-TAB: How do we interchange the data ?



MAGE-OM / MAGE-ML standards were complex for labs to adopt ...

So simpler MAGE-TAB standard developed based on spread-sheet TABular format

Sample ID	Characteristics [Organism]	Characteristics [OrganismPart]	Protocol REF	Extract ID	Protocol REF	Label	Hybridization ID	ArrayDesign REF	ArrayData URI	DerivedArrayData Matrix URI
liver 1	Homo sapiens	liver	P-XMPL-1	Extract 1	P-XMPL-3	Cy3	Hyb 1	SMD-10K	1.txt	FGDM.txt
liver 2	Homo sapiens	liver	P-XMPL-1	Extract 2	P-XMPL-3	Cy3	Hyb 1	SMD-10K	1.txt	FGDM.txt
kidney 1	Homo sapiens	kidney	P-XMPL-1	Extract 3	P-XMPL-3	Cy3	Hyb 2	SMD-10K	2.txt	FGDM.txt
kidney 2	Homo sapiens	kidney	P-XMPL-1	Extract 4	P-XMPL-3	Cy3	Hyb 2	SMD-10K	2.txt	FGDM.txt
brain 1	Homo sapiens	brain	P-XMPL-1	Extract 5	P-XMPL-3	Cy3	Hyb 3	SMD-10K	3.txt	FGDM.txt
brain 2	Homo sapiens	brain	P-XMPL-1	Extract 6	P-XMPL-3	Cy3	Hyb 3	SMD-10K	3.txt	FGDM.txt
				Extract reference	P-XMPL-3	Cy5	Hyb 1	SMD-10K	1.txt	FGDM.txt
				Extract reference	P-XMPL-3	Cy5	Hyb 2	SMD-10K	2.txt	FGDM.txt
				Extract reference	P-XMPL-3	Cy5	Hyb 3	SMD-10K	3.txt	FGDM.txt

Each 'layer' in the graph is represented by an ID column in the spreadsheet, followed by columns for each of the labels. Each path in the graph is represented by one row in the spreadsheet.

Public Microarray Database - ArrayExpress

The screenshot shows the ArrayExpress Home page in a Mozilla Firefox browser window. The address bar displays <http://www.ebi.ac.uk/microarray-as/ae/>. The page features a navigation bar with links to Databases, Tools, EBI Groups, Training, Industry, About Us, and Help. A search bar is prominently displayed with a "Go" button and a "Reset" link. Below the navigation bar, the main content area is divided into several sections:

- ARRAYEXPRESS**: The main header with the logo and a brief description of the database.
- Experiments Archive**: A section with 10199 experiments and 285018 assays. It includes a search bar for "Experiment, citation, sample and factor annotations" and a "Query" button.
- Gene Expression Atlas**: A section with 1255 experiments, 34264 assays, and 6532 conditions. It includes search bars for "Genes" and "Conditions", a "up/down in" dropdown, and a "Query" button.
- News**: A section with a recent update from 21 Dec 2009 regarding the ArrayExpress Bioconductor package.
- Links**: A section with various links including "ArrayExpress User Survey", "Help | Training | FAQ | Citing", "Submit Data", "Programmatic Access", "FTP Access", "Software Downloads and Statistics", "EFO | Bioconductor Package", "Quality Metrics", "ArrayExpress Scientific Advisory Board", and "Microarray Informatics Group".

At the bottom of the page, there is a footer with links to "Terms of Use", "EBI Funding", and "Contact EBI". A search bar is also present at the bottom with a "Find:" label and a "Match case" checkbox. The Zotero logo is visible in the bottom right corner.

This is entirely based around the MAGE-OM model, with an SQL database automatically generated from the object model standard and accepts submission in MAGE-ML format

Public Microarray Database – GEO

The screenshot shows the Gene Expression Omnibus (GEO) Main page in Mozilla Firefox. The browser's address bar displays the URL <http://www.ncbi.nlm.nih.gov/geo/>. The page features the NCBI logo and the GEO logo. A navigation bar includes links for HOME, SEARCH, SITE MAP, GEO Publications, FAQ, MIAME, and Email GEO. The main content area is titled "NCBI » GEO" and includes a login status "Not logged in | Login". A description of the Gene Expression Omnibus is provided, stating it is a public functional genomics data repository supporting MIAME-compliant data submissions. The page is divided into two main sections: "GEO navigation" and "Site contents".

GEO navigation

- QUERY**
 - DataSet:
 - Gene profiles:
 - GEO accession:
 - GEO BLAST
- BROWSE**
 - DataSet
 - Platforms
 - GEO accessions
 - Samples
 - Series

Site contents

- Public data**

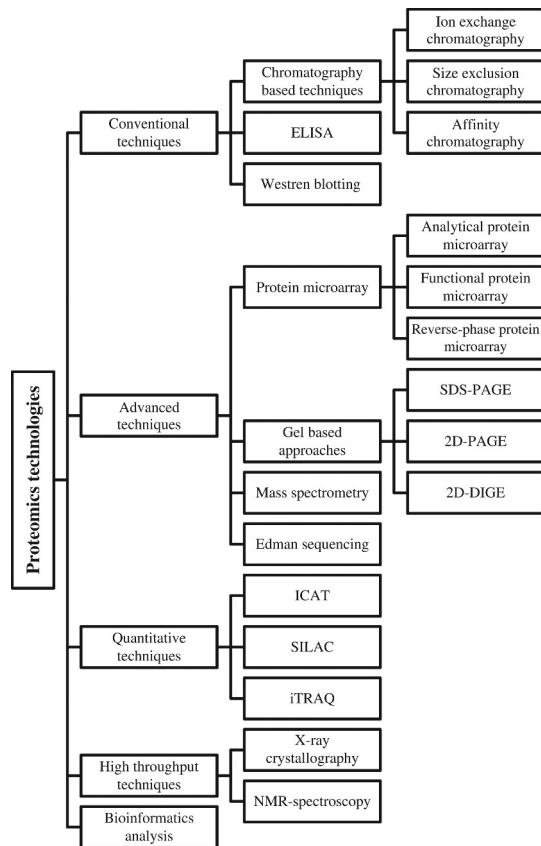
Platforms	7,132
Samples	414,936
Series	16,155
- Documentation**
 - Overview | FAQ | Find
 - Submission guide
 - Linking & citing
 - Journal citations
 - Programmatic access
 - DataSet clusters
 - GEO announce list
 - Data disclaimer
 - GEO staff

The bottom of the page features a search bar with the text "Find:" and a "Done" button. Navigation links for "Previous", "Next", "Highlight all", and "Match case" are also present. The Zotero logo is visible in the bottom right corner.

Based on a simpler data model and does **not** encourage MAGE-ML submission.

But does encourage MIAME-compatible data and MAGE-TAB.

Proteomics. What do we need to capture?



Experimental Metadata

Mass spectrometry results/design

Protein separation experiments

Molecular Interactions

HUPO: Proteomics Standards Initiative

Since 2002

12 Data Modelling standards/guidelines

mass spec, interactions, protein affinities

18 Data storage formats

5 controlled vocabularies

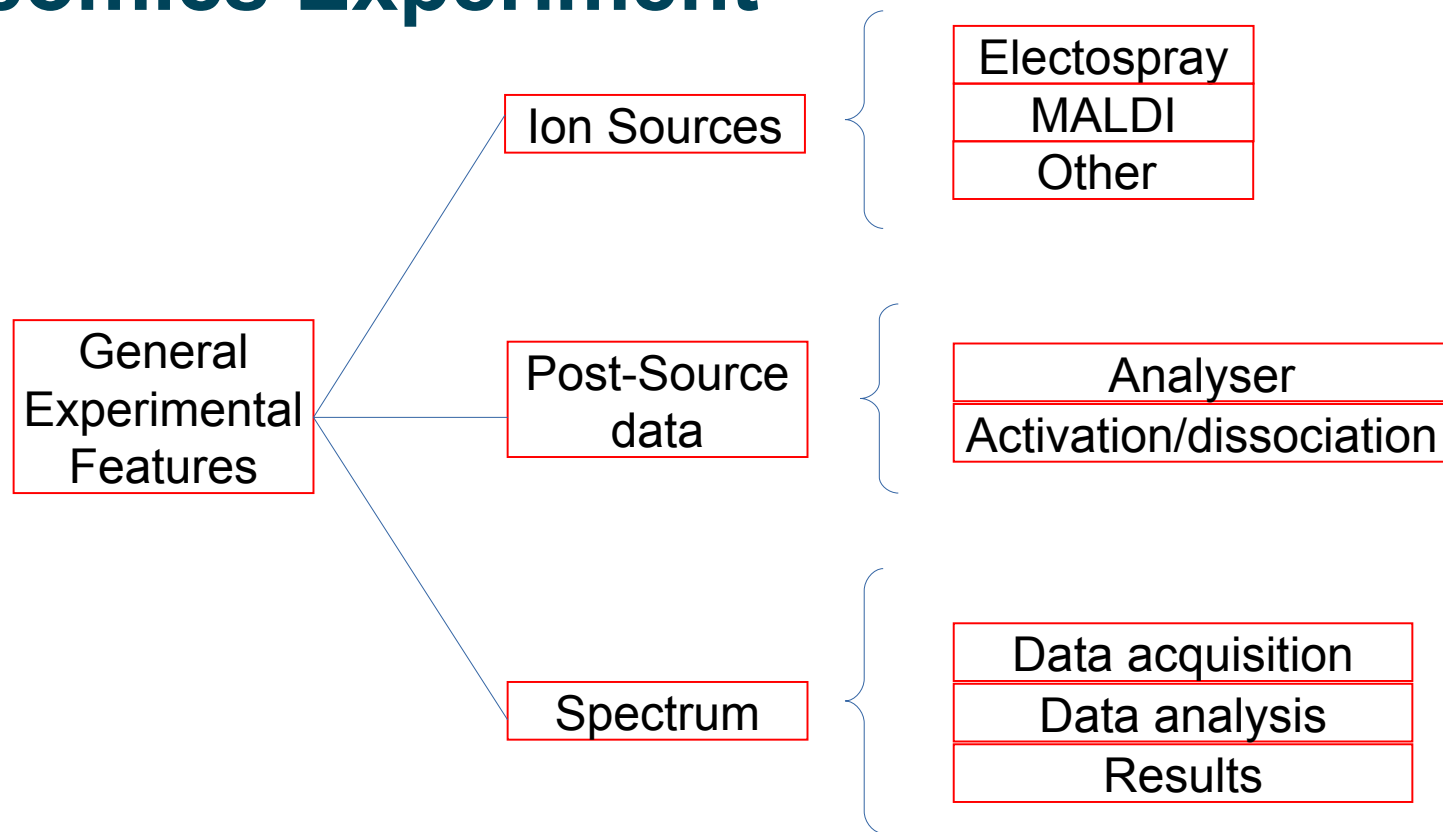
<http://www.psidev.info>

<https://github.com/HUPO-PSI>

MIAPE-XX: Minimum Information About a Proteomics Experiment

MODULE	DATA
MIAPE-MS	Mass Spectrometry
MIAPE-MSI	Mass Spectrometry Informatics
MIAPE-Quant	Mass Spectrometry Quantification
MIAPE-GE	Gel Electrophoresis
MIAPE-GI	Gel Informatics
MIAPE-CC	Column Chromatography
MIAPE-CE	Capillary Electrophoresis
MIMix	Molecular Interactions

MIAPE-MS: Minimum Information About a Proteomics Experiment



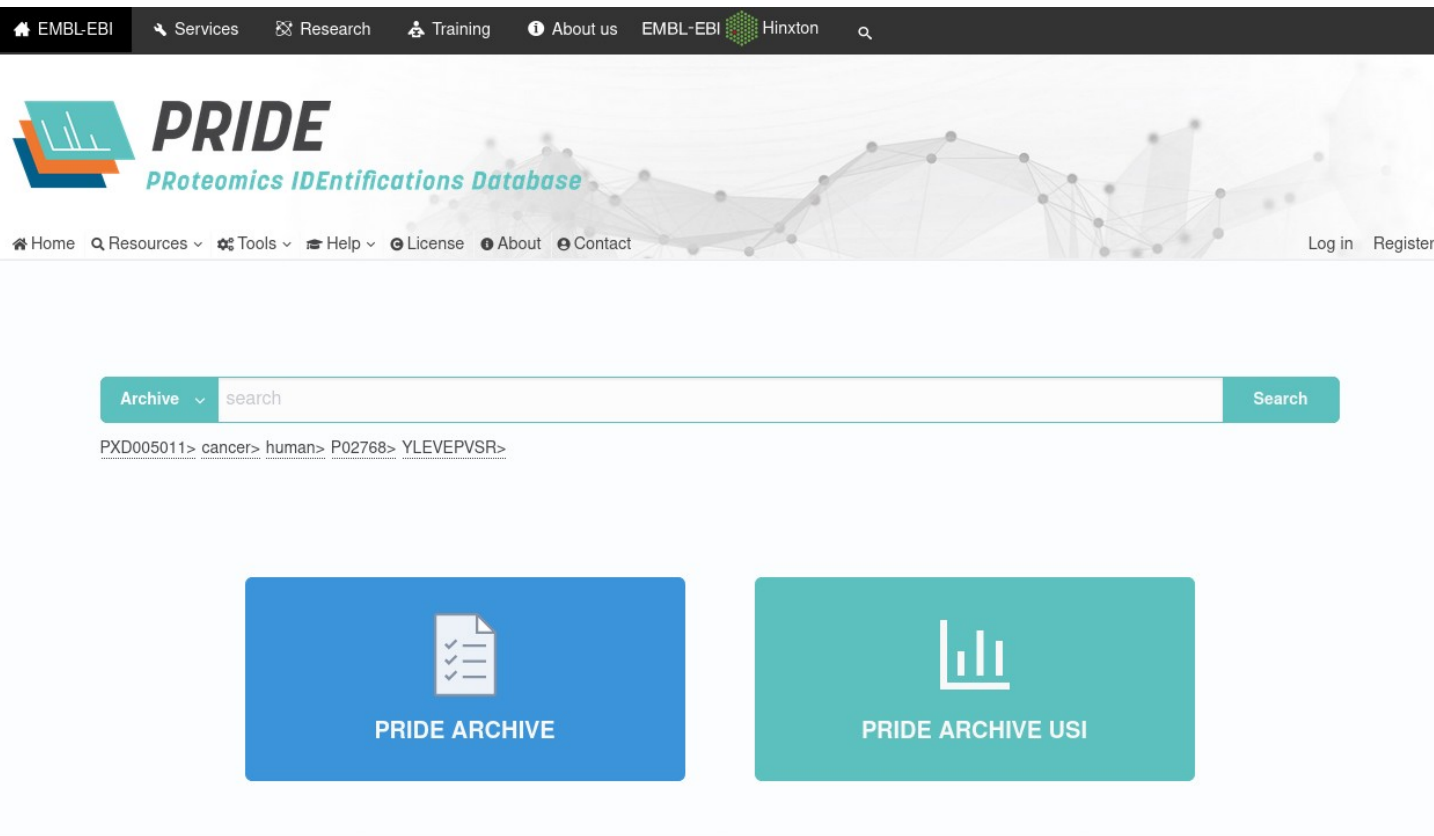
mzML: Mass Spectrometry Markup Language

Fuses together two prior standards; mzData and mzXML

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-<dataProcessing>
  -<software>
    <name>Bioworks Browser</name>
    <version>3.4 Build 3</version>
  </software>
  -<processingMethod>
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    <cvParam cvLabel="psi" accession="PSI:1000034" name="ChargeDeconvolutionW" value="false"/>
    <cvParam cvLabel="psi" accession="PSI:1000035" name="PeakProcessing" value="discrete"/>
  </processingMethod>
</dataProcessing>
```

PRIDE: PRoteomics IDentification Database

MIAPE compliant
data resource at
the EMBL-EBI



The screenshot shows the PRIDE website interface. At the top is a dark navigation bar with links for EMBL-EBI, Services, Research, Training, About us, and EMBL-EBI Hinxton. Below this is the PRIDE logo and a network diagram. A secondary navigation bar contains links for Home, Resources, Tools, Help, License, About, and Contact, along with Log in and Register buttons. A search bar is present with a dropdown menu set to 'Archive' and a 'Search' button. Below the search bar, a breadcrumb trail reads 'PXD005011> cancer> human> P02768> YLEVEPVSR>'. Two large buttons are displayed: 'PRIDE ARCHIVE' (blue) and 'PRIDE ARCHIVE USI' (teal). At the bottom, there are three small sections: 'Submitted datasets properties tree', 'Number of Submissions', and 'Submitted datasets per country', each with a 'More' link.

EMBL-EBI Services Research Training About us EMBL-EBI Hinxton

PRIDE
PRoteomics IDentifications Database

Home Resources Tools Help License About Contact Log in Register

Archive search Search

PXD005011> cancer> human> P02768> YLEVEPVSR>

PRIDE ARCHIVE

PRIDE ARCHIVE USI

Submitted datasets properties tree More

Number of Submissions More

Submitted datasets per country More

Metabolomics. What do we need to capture?

Sample preparation details

Minimum set of data for chromatography, Mass Spectrometry, small molecule NMR, FT-IR spectroscopy

Minimum set of data for stable isotopes & flux analysis

Instrument performance and validation

Data preprocessing steps

Metabolite identification

Metabolomics Standards Initiative

Since 2005, standard since 2007

The Metabolomics Society – Data Standards Task Group

Metabolic Standards Initiative (MSI)

Core Information For Metabolomics Reports (CIMR)

<https://github.com/MSI-Metabolomics-Standards-Initiative/>

Compliance?

MetaboLights, Metabolomics Workbench, Metabolomics Repository Bordeaux (MeRy-B) and Metabolic Phenotype Database (MetaPhen)

Deposited database compliance: patchy 0 to 97%

No Journal requirements



MetaboLights

MetaboLights is a database for Metabolomics experiments and derived information. The database is cross-species, cross-technique and covers metabolite structures and their reference spectra as well as their biological roles, locations and concentrations, and experimental data from metabolic experiments. MetaboLights is the recommended Metabolomics repository for a number of leading journals.

[More about us](#)[Quick tour](#) ➔[Tweets by @MetaboLights](#)

Study

[BROWSE](#)[ORCID SEARCH](#)

Compound Library

[COMPOUNDS](#)[SPECIES](#)

Training

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MIAPE compliant
data resource @
EMBL-EBI

MetaboLights is the recommended Metabolomics repository for a number of leading journals.

[SCIENTIFIC DATA](#)[BioMed Central](#)[PLOS BIOLOGY](#)[EMBOpress](#)[F1000Research](#)[METABOLOMICS](#)



METABOLOMICS WORKBENCH

Log in / Register

Search the Metabolomics Workbench

Home | Data Repository | Databases | Protocols | Tools | About | Search

Welcome to the UCSD Metabolomics Workbench, a resource sponsored by the Common Fund of the National Institutes of Health.

National Metabolomics Data Repository

[Upload and Manage Studies](#)
[Browse and Search Studies](#)
[Analyze Studies](#)

As of **11/10/23** a total of **2887 studies** have been processed by the National Metabolomics Data Repository (NMDR). There are **2533 publicly available studies** and the remainder (354) will be made available subject to their embargo dates.

Recently released studies on NMDR

ST002404 - Profiling Metabolites and Lipoproteins in COMETA, an Italian Cohort of COVID-19 patients-part 2; *Homo sapiens*; [University of Florence](#)

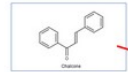
ST002729 - Improved Endurance Capacity of Diabetic Mice during SGLT2 Inhibition: Potential Role of AICARP, an Endogenous AMPK Activator.; *Mus musculus*; [Medical Institute of Bioregulation, Kyushu University](#)

ST002928 - The major TMEM106B dementia risk allele affects TMEM106B protein levels, fibril formation, and myelin lipid homeostasis in the ageing human hippocampus; *Homo sapiens*; [University of Sydney](#)

Core structures in RefMet classification

RefMet core structures (March 1, 2023)

[Browse](#) and search core structures associated with the [RefMet classification system](#). For example, what are "Flavones", "Flavanols", "Flavonols" and "Flavanones" and what's the difference? The RefMet classification hierarchy has recently been updated to place more emphasis on biosynthetic considerations for Alkaloid, Polyketide and Prenol lipid super classes.



RefMet core structures					
A database of core structures representing sub-classes in the RefMet classification system					
Enter all structure search					
Name (Contains)	Super class	Super class	Sub class	Formula	Count
Chalcone	Polyketide	Flavonoid	Chalcone	C ₁₅ H ₁₀ O ₂	207,000
Flavone	Polyketide	Flavonoid	Flavone	C ₁₅ H ₁₀ O ₂	208,000
Flavonol	Polyketide	Flavonoid	Flavonol	C ₁₅ H ₁₀ O ₃	184,000
Flavonol	Polyketide	Flavonoid	Flavonol	C ₁₅ H ₁₀ O ₃	140,000
Flavonol	Polyketide	Flavonoid	Flavonol	C ₁₅ H ₁₀ O ₃	226,000
Flavonol	Polyketide	Flavonoid	Flavonol	C ₁₅ H ₁₀ O ₃	220,000

Quick Links - Key Resources

[Follow @MetabolomicsWB](#) [Tweets by MetabolomicsWB](#)

NIH Common Fund Stage 2 Metabolomics Consortium Centers

Metabolomics Consortium Coordinating Center (M3C)
Richard Yost, U. of Florida

[Metabolomics Workbench/NMDR](#)
Shankar Subramaniam, UC San Diego (this website)

Compound Identification Cores (CIDs)
Arthur Edison, U. of Georgia
Alexey Nesvizhskii, U. of Michigan
Oliver Fiehn, UC Davis
Dean Paul Jones, Emory University
Thomas Metz, Pacific Northwest Nat. Lab.

Data and Tools Cores (DTCs)
John Weinstein, MD Anderson Cancer C.
Jamey Young, Vanderbilt University
Xiuxia Du, U. of North Carolina Charlotte
Shuzhao Li, Emory University
Alla Karnovsky, U. of Michigan
Katerina Kechris, U. of Colorado, Denver
Gary Patti, Washington U. at St. Louis

Please cite: [Metabolomics Workbench](#) [You will get more info on how to cite here](#)

UCSD hosted

Orthogonal Data Standards

Gene Ontology

“The mission of the GO Consortium is to develop a comprehensive, computational model of biological systems, ranging from the molecular to the organism level, across the multiplicity of species in the tree of life.” - circa 2023

Circa 1998

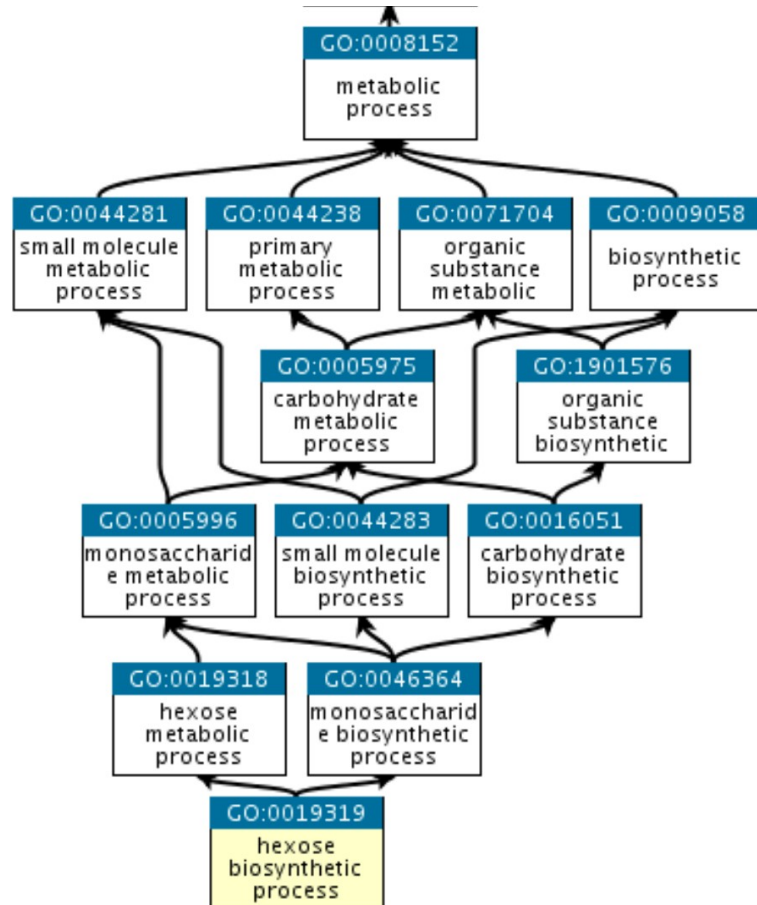
<https://geneontology.org/>

Gene Ontology

A formal ontology to describe the function of gene products

Ontology	Description
Molecular Function	Molecular level activities performed by gene products
Cellular Component	Location of gene products, relative to cellular components and structures
Biological Process	Larger cellular process the gene products is involved in

Biological Process



3 Directed Acyclic Graphs

Gene products annotated with terms “implies” parent terms

AmiGO 2


AmiGO 2
Home
Search ▾
Browse
Tools & Resources
Help
Feedback
About

Quick search

Search ?

hexose biosynthetic process

Term Information ?

Accession GO:0019319

Name hexose biosynthetic process Data health ♥

Ontology biological_process

Synonyms hexose anabolism, hexose biosynthesis, hexose formation, hexose synthesis

Alternate IDs None

Definition The chemical reactions and pathways resulting in the formation of hexose, any monosaccharide with a chain of six carbon atoms in the molecule. *Source:* ISBN:0198506732

Comment None

History See term [history](#) for GO:0019319 at QuickGO

Subset None

Related

- [Link](#) to all **genes and gene products** annotated to hexose biosynthetic process (**excluding "regulates"**).
- [Link](#) to all direct and indirect **annotations** to hexose biosynthetic process (**excluding "regulates"**).
- [Link](#) to all direct and indirect **annotations download** (limited to first 10,000) for hexose biosynthetic process (**excluding "regulates"**).

[Include "regulates"](#)

For more information, please see the [ontology relation documentation](#).

Annotations
Graph Views
Inferred Tree View
Neighborhood
Mappings

Filter results

Total annotations: 2496

User filters
+ isa_partof_closure: GO:0019319

Total annotations: 2496; showing: 1-10
Results count 10 ▾

«First»
«Prev»
Next»
Last»
Download

☐	Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence
☐	GL50803_00113021	Acetyl-CoA carboxylase/pyruvate carboxylase fusion protein, putative		gluconeogenesis		GO_Central	Giardia lamblia ATCC 50803	IBA	PANTHER SGD:S00
☐	Os04g0234600	Os04g0234600 protein		gluconeogenesis		GO_Central	Oryza sativa Japonica	IBA	PANTHER RGD:259



BLAST

Align

Peptide search

ID mapping

SPARQL

UniProtKB

Advanced | List

Search



Help

Function

Names & Taxonomy

Subcellular Location

Disease & Variants

PTM/Processing

Expression

Interaction

Structure

Family & Domains

Sequence & Isoforms

Similar Proteins

Entry

Variant viewer 213

Feature viewer

Genomic coordinates new

Publications

External links

Expand table

History

GO annotations¹

Slimming set:

generic



Cell color indicative of number of GO terms

ASPECT TERM

Cellular
Component

apical part of cell

Source:Ensembl

Cellular
Component

axon

Source:UniProtKB

Cellular
Component

cell surface

Source:UniProtKB

2 Publications

Cellular
Component

cell-cell junction

Source:Ensembl

Cellular
Component

ciliary rootlet

Source:Ensembl

Cellular
Component

clathrin-coated pit

Source:UniProtKB-SubCell

Cellular

COPII-coated ER to Golgi transport vesicle

Source:Ensembl

Feedback

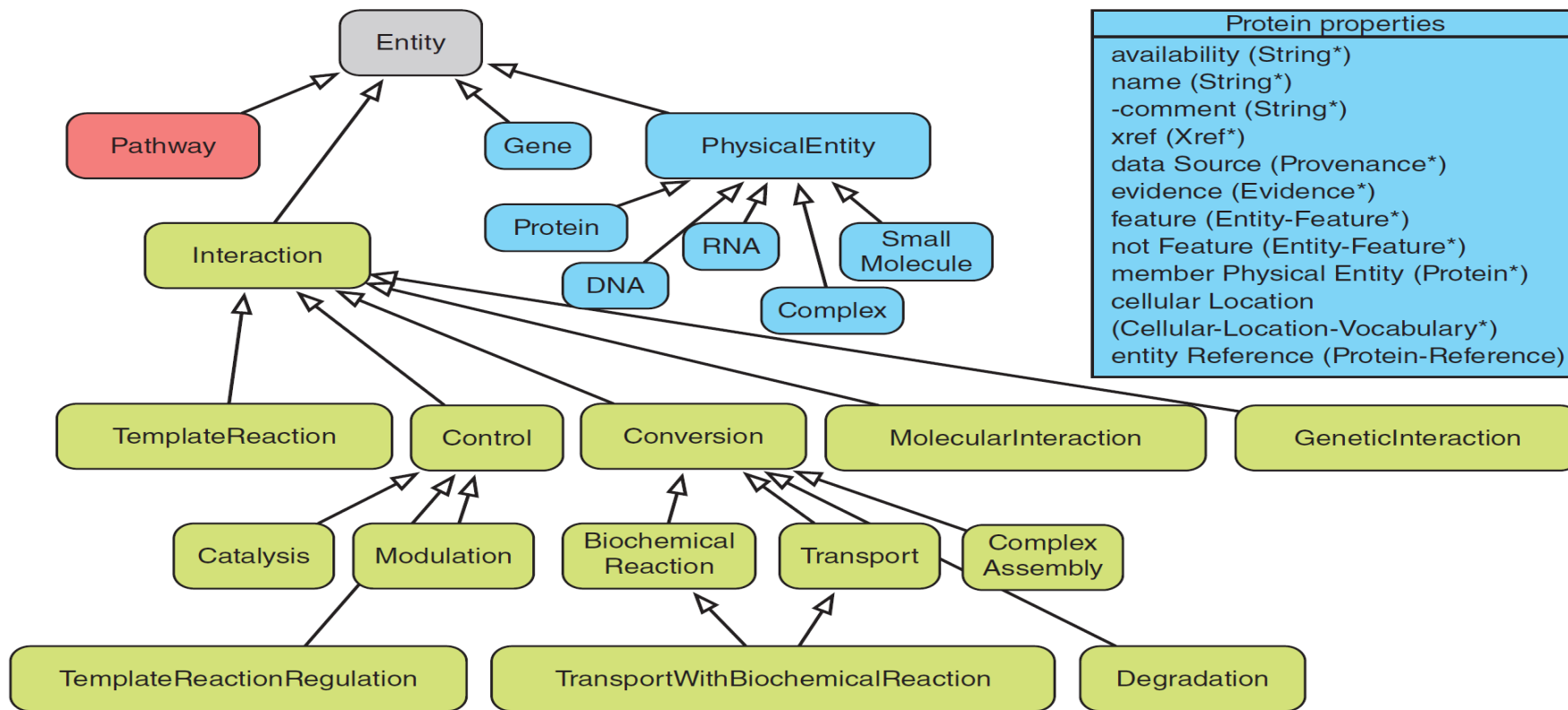
Help

Expand table

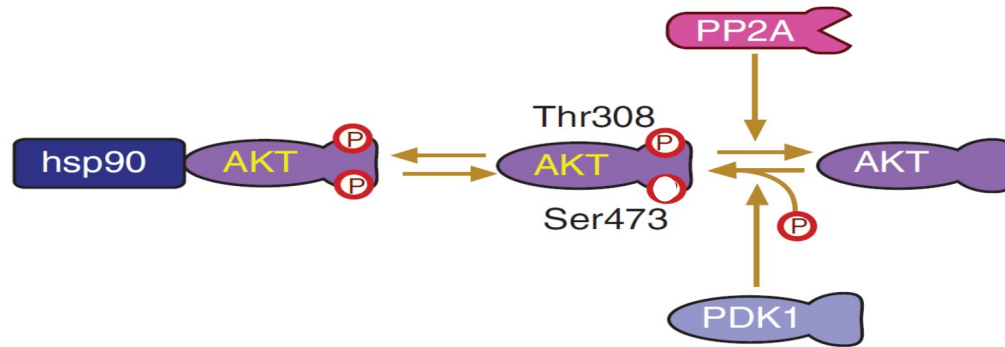
BioPAX language: a formal representation of biological pathway knowledge

- You have seen the Gene Ontology which allows machine reasoning about gene function.
- The BioPAX language uses similar ideas to represent biological pathway concepts so that they can be reasoned about.
- The BioPAX language uses the OWL ontology language which is a W3 XML standard developed for the “semantic web”.
- In addition to the BioPAX ontology, the Systems Biology Graphical Notation” (SBGN) allows human comprehension of pathway information.

High-level overview of BioPAX classes



In more concrete terms – representing one part of the AKT signaling pathway

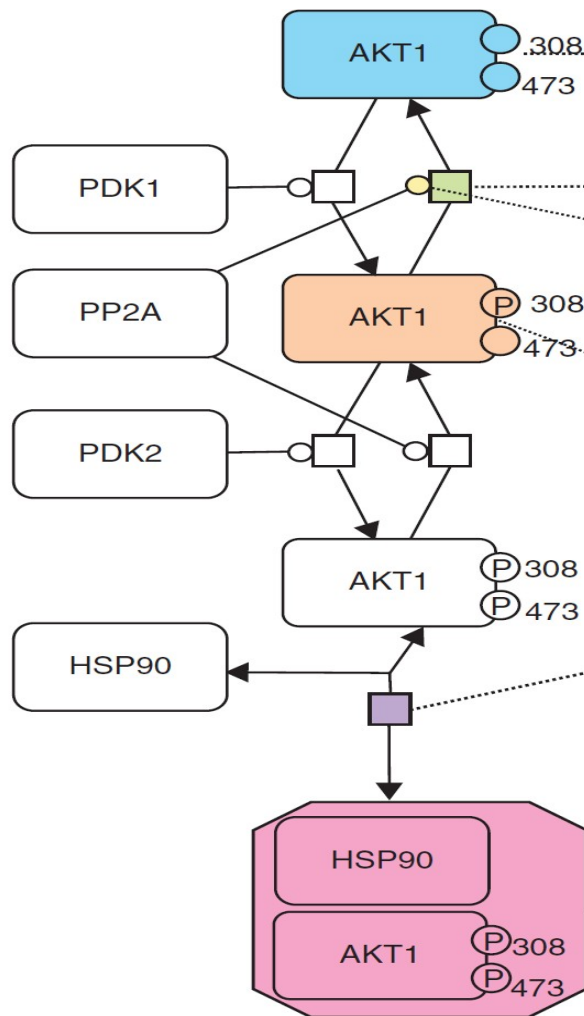


rAKT1 is a *ProteinReference*
has *standard-name* "AKT1"
has *name* "PKB"
has *xref* Uniprot-P31749

p@308 is a *ModificationFeature*
has *featureLocation* AKT1-308
has *modificationType*
phosphorylation

SBGN / BioPAX representation of AKT pathway

Allows formal
representation
of the biology



AKT1.1 is a *Protein*
has *proteinReference* rAKT1
has *notFeature* p@308
has *notFeature* p@473

reaction1 is a *BiochemicalReaction*
has *left* AKT1.2
has *right* AKT1.1
is *left-to-right*.

catalysis1 is a *Catalysis*
has *controller* PP2A.1
has *controlled* reaction1
has *direction* irr-left-to-right

AKT1.2 is a *Protein*
has *proteinReference* rAKT1
has *feature* p@308
has *notFeature* p@473

assembly1 is a *ComplexAssembly*
has *left* HSP90.1
has *left* AKT1.3
has *right* complex1
is *reversible*

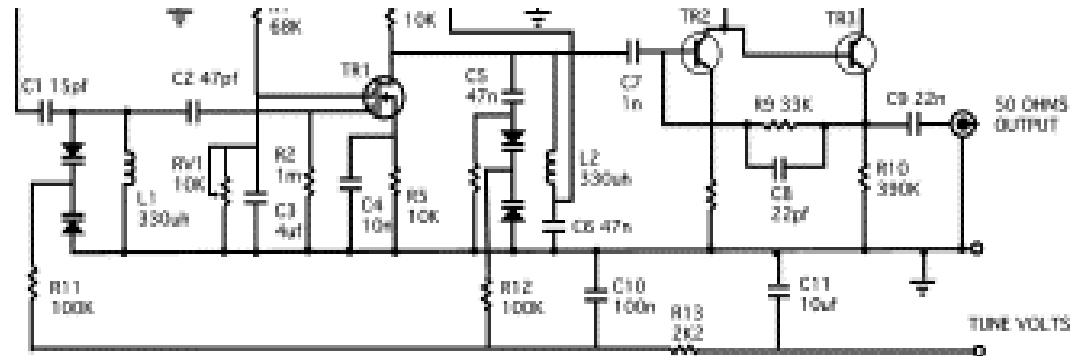
complex1 is a *Complex*
has *component* AKT1.4
has *component* HSP90.2

HSP90.2 is a *Protein*
has *proteinReference* rHSP90
is *boundTo* AKT1.4

AKT1.4 is a *Protein*
has *proteinReference* rAKT1
has *feature* p@308
has *feature* p@473
is *boundTo* HSP90.2

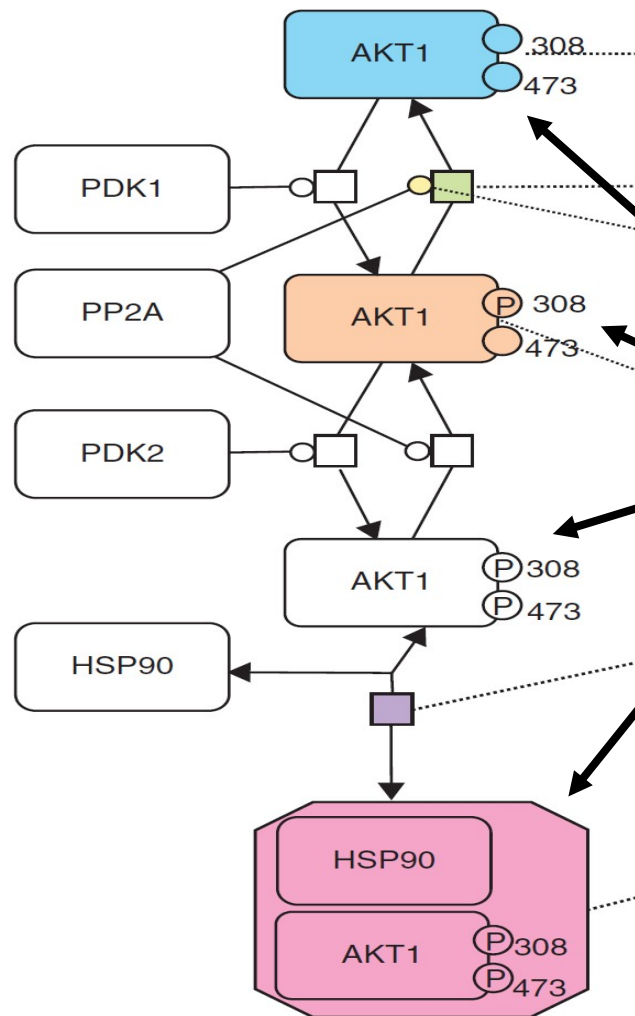
How would we use such pathway knowledge to help integrate data and understand biological systems?

- Using a radio analogy for inspiration ... we could measure voltages throughout the circuit between a working radio and the broken one.
- Essentially overlaying relevant 'high throughput' data to try to pinpoint faulty components in the circuit.



The role of -omics data

Proteomics data (or gene expression data as a “rough proxy”) for enzyme levels (or essentially enzyme activity).



Proteomics data for different levels of phosphorylated proteins and complexes present in the samples.

In Summary

- We briefly summarised different types of -omics experiments
- Briefly looked at data standards for -omics experiments
- Mentioned the different data resources
- Had a look at the meta-data standards GO and BioPAX